GLOBAL EDITION

MICROECONOMICS



Ch 2

Qu:
- 1-9
- 11-14
18/19

(Quiz will be in MCQ form)

14th Edition

graph - understand PPF

calculations - opp costs / comparative / absolute advantage

terms - difference - understand directly

Michael Parkin

(↳ which topic falls under which type of economic branch

↳ addresses

Macro → national economy

Micro → any individual behavior





2 THE ECONOMIC PROBLEM

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After studying this chapter, you will be able to:

- Define the production possibilities frontier and use it to calculate opportunity cost
- Define preferences and marginal benefit and describe an efficient allocation of resources
- Explain how specialization and trade make resource use more efficient
- Explain how current production choices expand and change future production possibilities
- Describe the economic institutions that coordinate decisions

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Production Possibilities and Opportunity Cost

The **production possibilities frontier** (*PPF*) is the boundary between those combinations of goods and services that can be produced and those that cannot.

To illustrate the *PPF*, we focus on two goods and hold the quantities of all other goods and services constant.

That is, we look at a model economy in which everything remains the same (*ceteris paribus*) except the two goods we're considering.

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PPF is concave-shaped

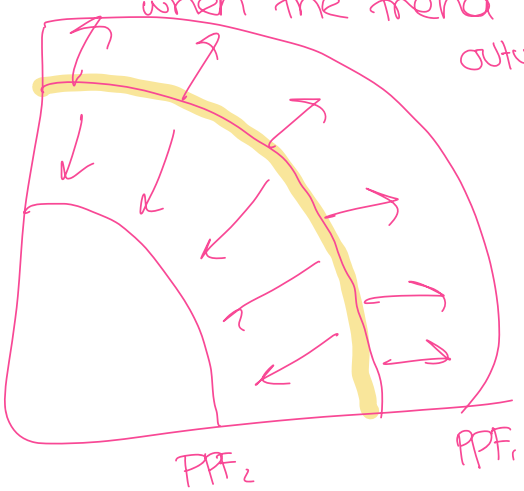




→ convex

↗ leads to higher/inc rates of employment

→ more possibilities/production when the trend of the PPF moves outward, resulting in economic growth in the country



↓ concave illustrates the increased opp. costs

you give up 2 units of food to gain 1 car

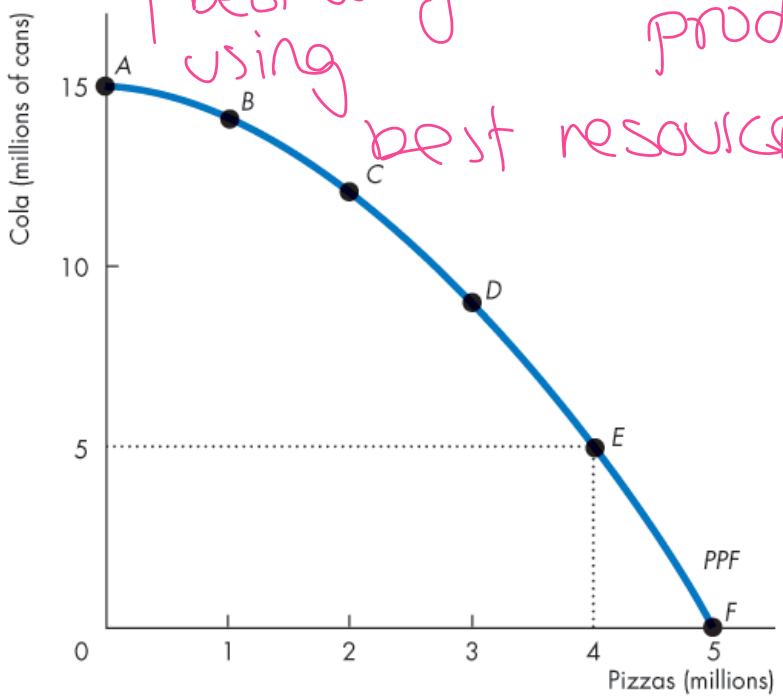
→ opp cost of 1 car is 2 units of food

Production Possibilities and Opportunity Cost

Production Possibilities Frontier

Figure 2.1 shows the *PPF* for two goods: cola and pizzas.

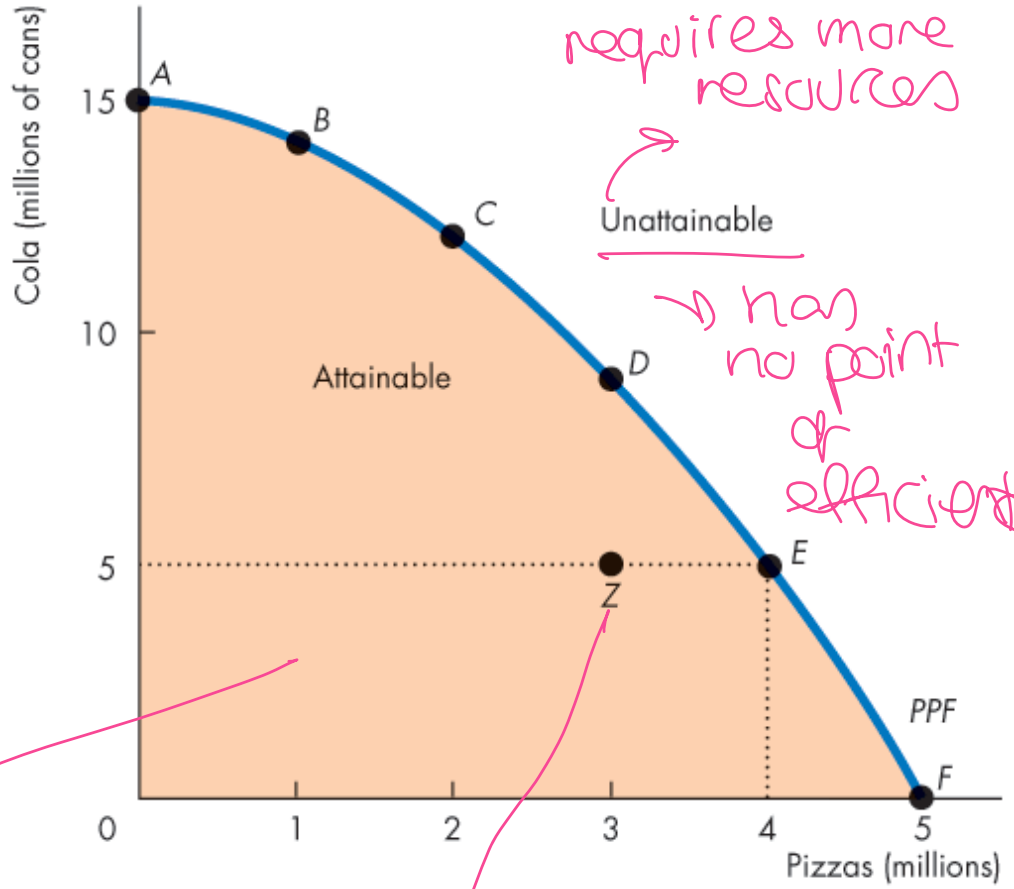
Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0



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- any point on the curve = attainable + efficient
- any point inside the curve = attainable + inefficient
- any point outside the curve = unattainable

Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0



requires more resources

↳ has no point of efficiency

results in misallocation → unemployment

attainable but it is not efficient bc

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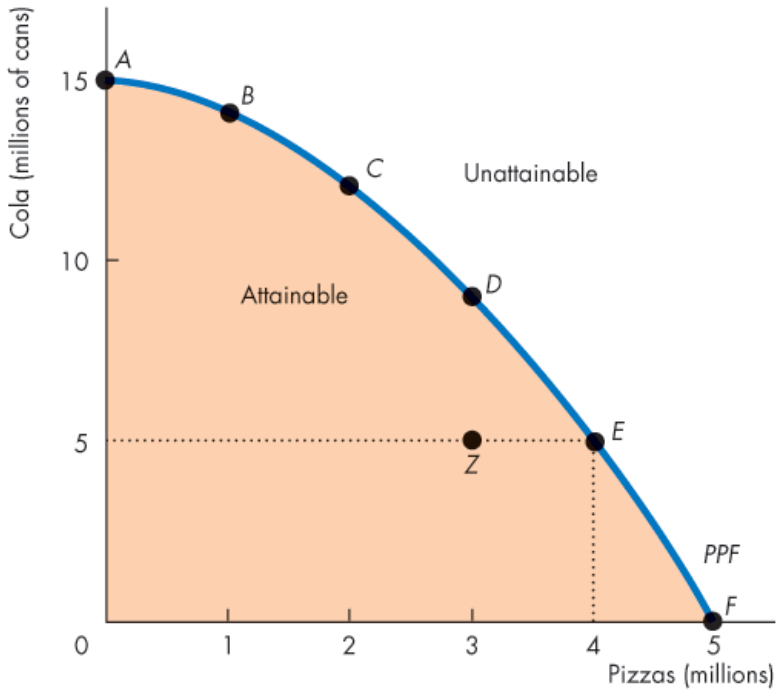
Production Possibilities and Opportunity Cost

it is not utilizing ALL the resources

Any point *on* the frontier such as *E* and any point *inside* the *PPF* such as *Z* are attainable.

Points outside the *PPF* are unattainable.

Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0



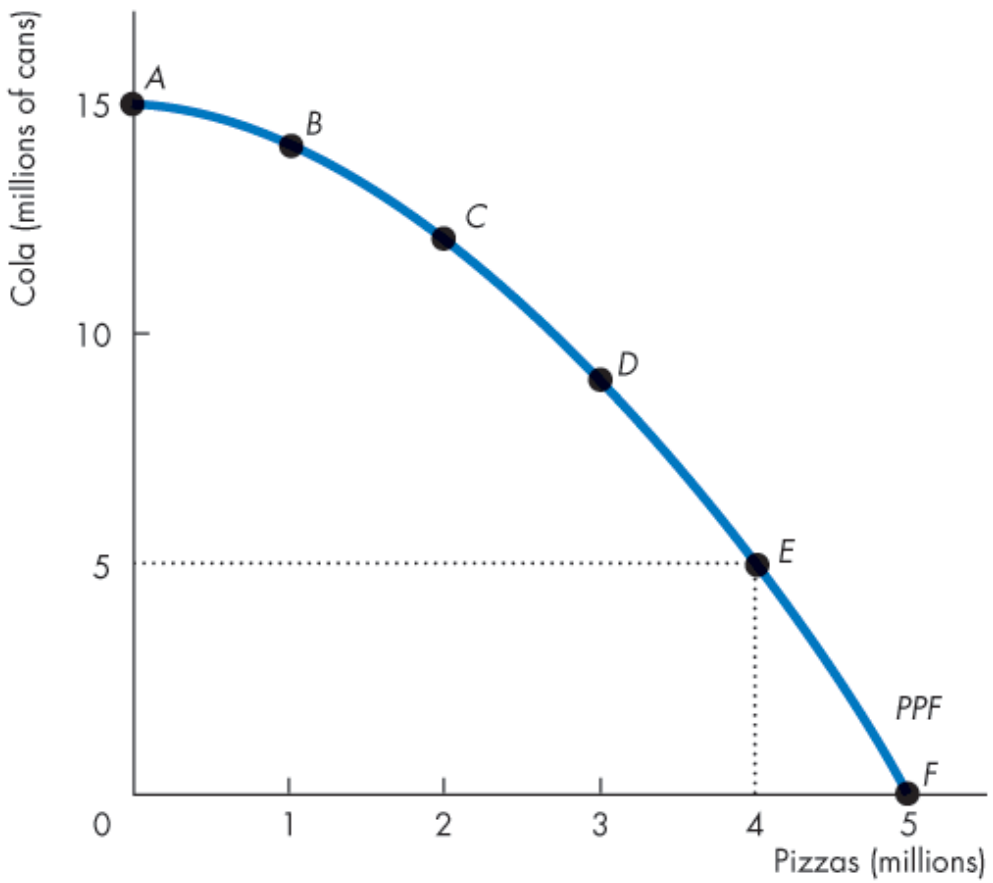
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Production Possibilities and Opportunity Cost

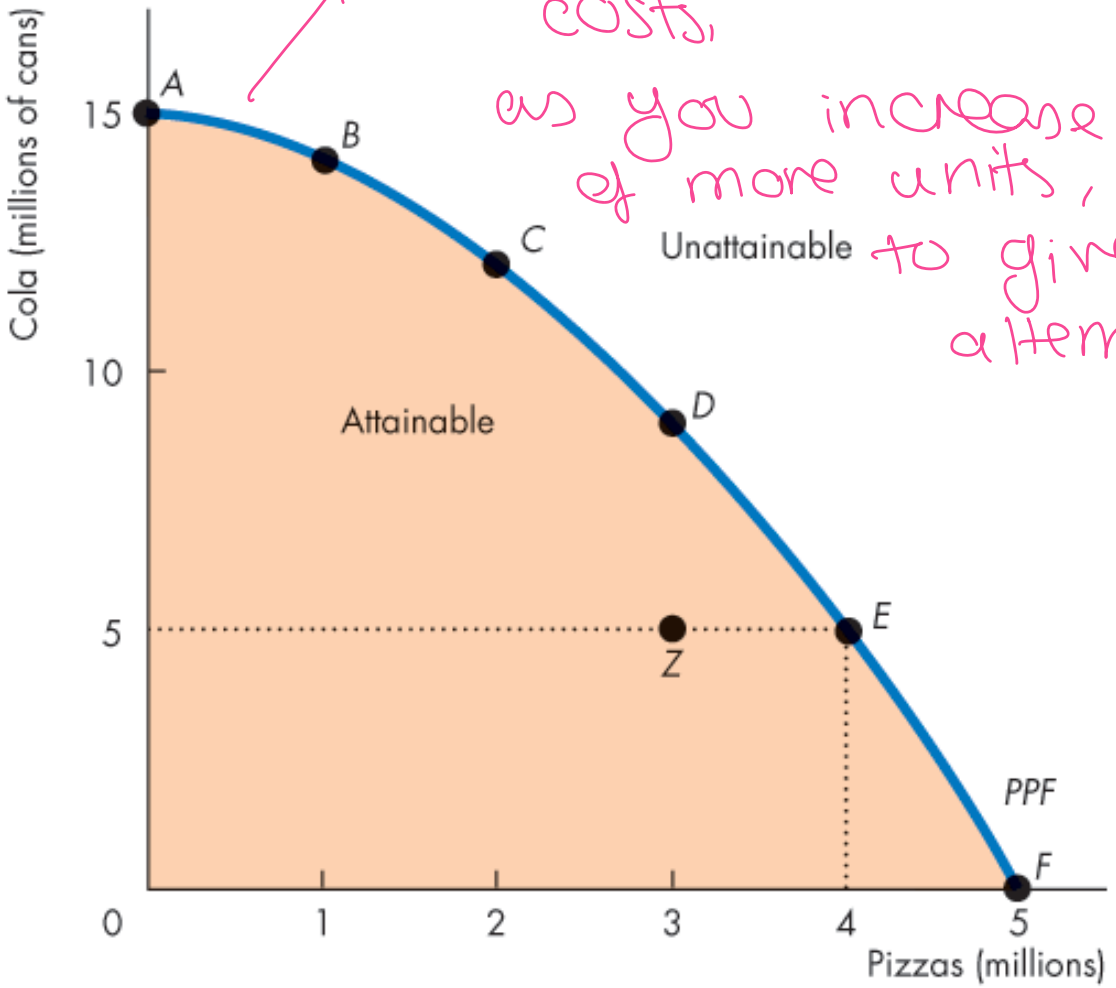
Production Efficiency

We achieve **production efficiency** if we cannot produce more of one good without producing less of some other good.

All points on the *PPF* are *efficient*.



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shows increased opportunity costs.

as you increase the production of more units, you will have to give up on the alternative

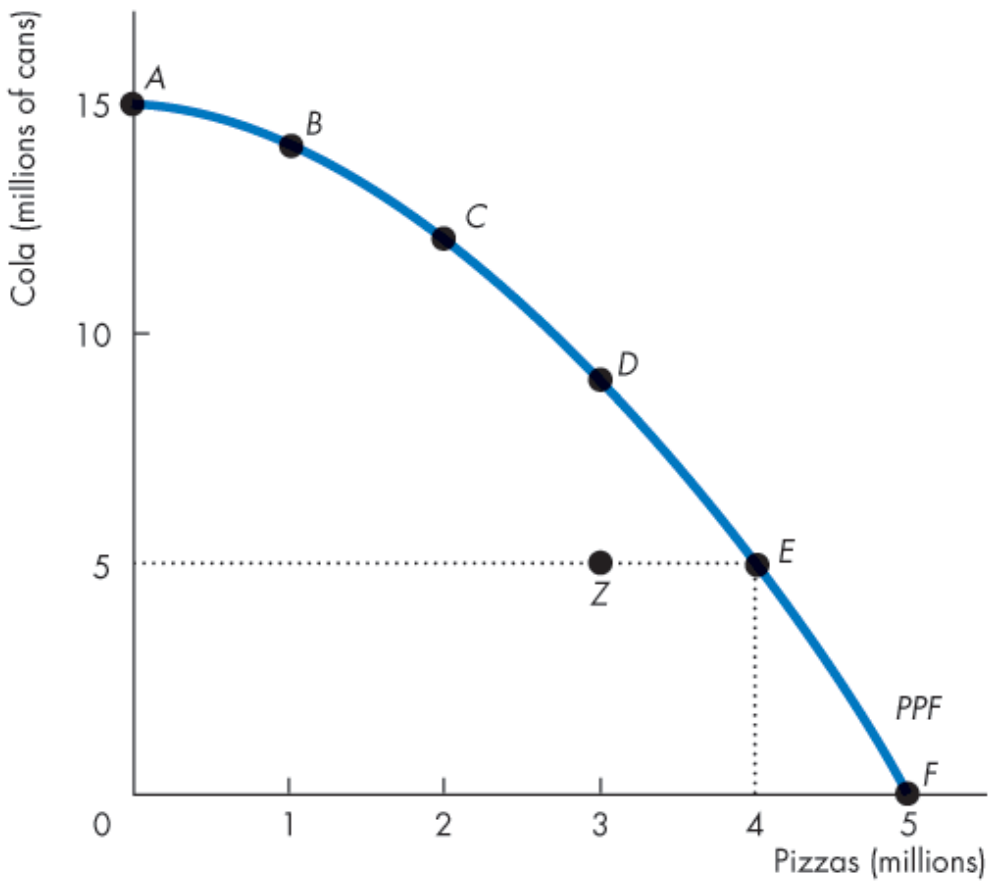
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Production Possibilities and Opportunity Cost

Any point inside the frontier, such as *Z*, is *inefficient*.

At such a point, it is possible to produce more of one good without producing less of the other good.

At *Z*, resources are either unemployed or misallocated.



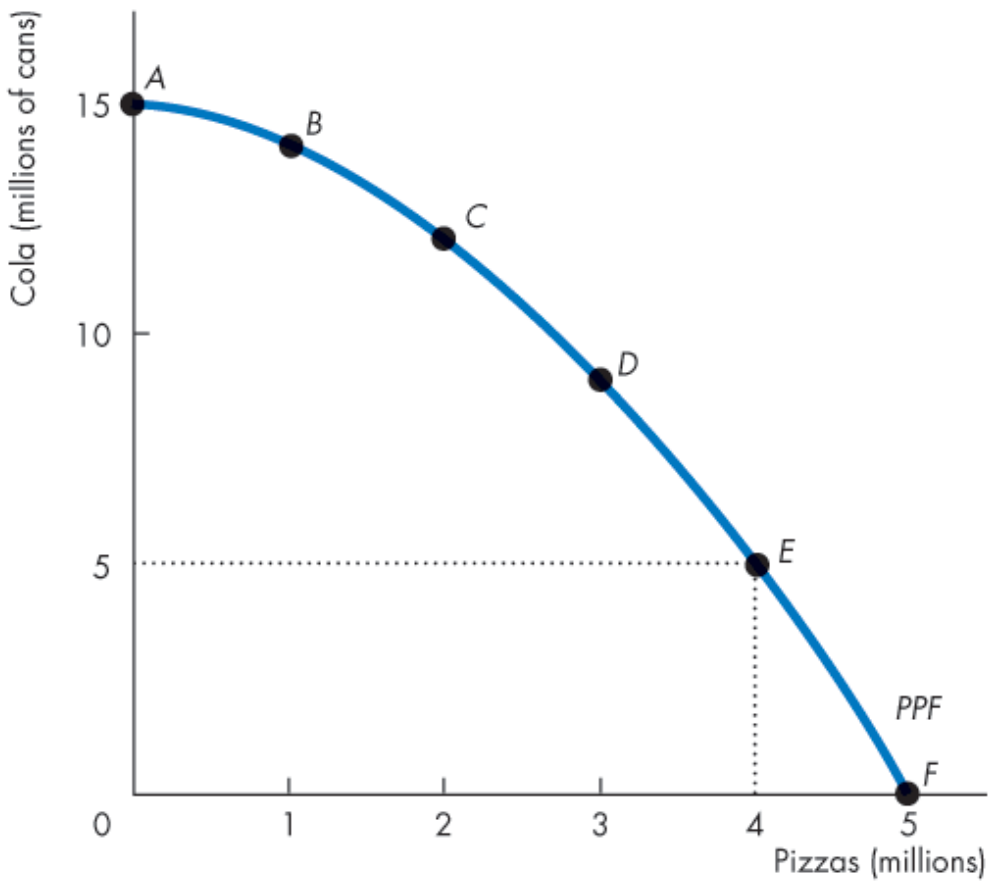
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Production Possibilities and Opportunity Cost

Tradeoff Along the *PPF*

Every choice along the *PPF* involves a *tradeoff*.

On this *PPF*, we must give up some cola to get more pizzas or we must give up some pizzas to get more cola.



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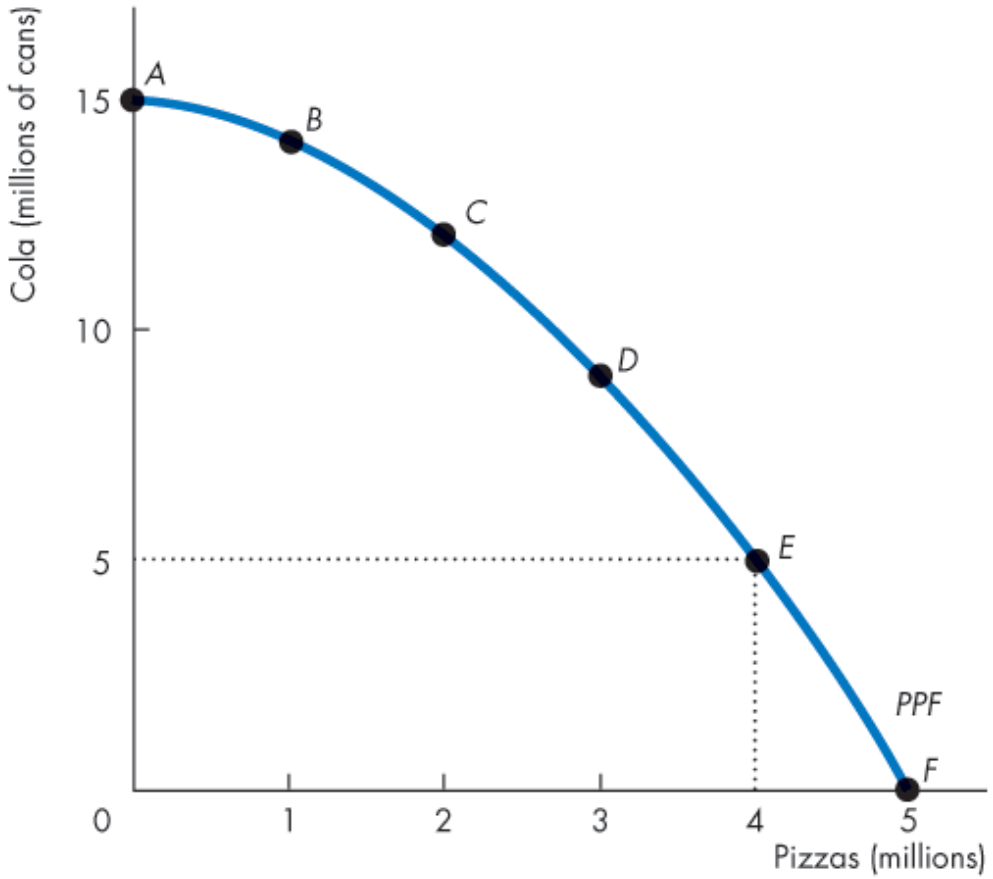
Production Possibilities and Opportunity Cost

Opportunity Cost

As we move down along the *PPF*,

we produce more pizzas, but the quantity of cola we can produce decreases.

The opportunity cost of a pizza is the cola forgone.



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Production Possibilities and Opportunity Cost

there is an opportunity cost of

In moving from *E* to *F*:

The quantity of pizzas increases by 1 million.

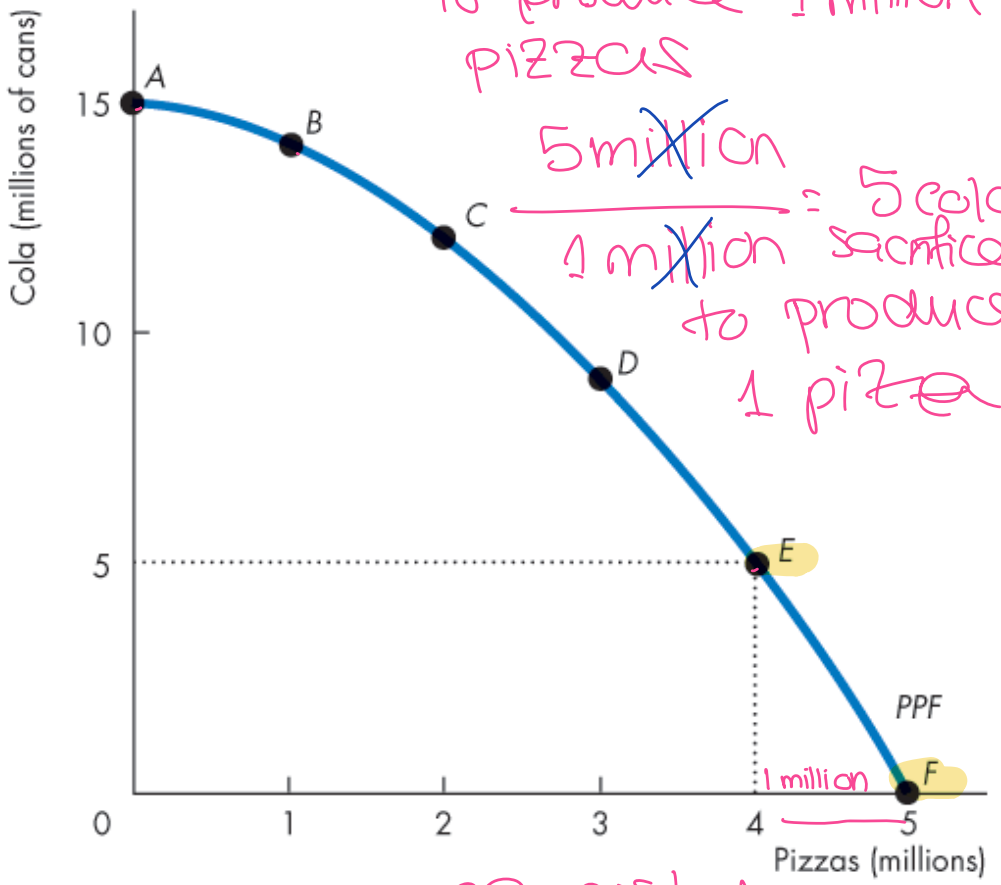
The quantity of cola decreases by 5 million cans.

The opportunity cost of the fifth 1 million pizzas is 5 million cans of cola.

One of these pizzas costs 5 cans of cola.

5 million cola to produce 1 million pizzas

$\frac{5 \text{ million}}{1 \text{ million}} = 5$ colas sacrificed to produce 1 pizza



opp cost of one pen = $\frac{\text{give up}}{\text{gain}} \rightarrow \frac{6 \text{ pencil}}{2 \text{ pens}} = 3 \text{ pencils}$

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↳ inverse if asked about another product

Production Possibilities and Opportunity Cost

in order to produce 5 million colas, we must give up on 1 million pizza

1 million / 5 million = cost of producing 1 cola

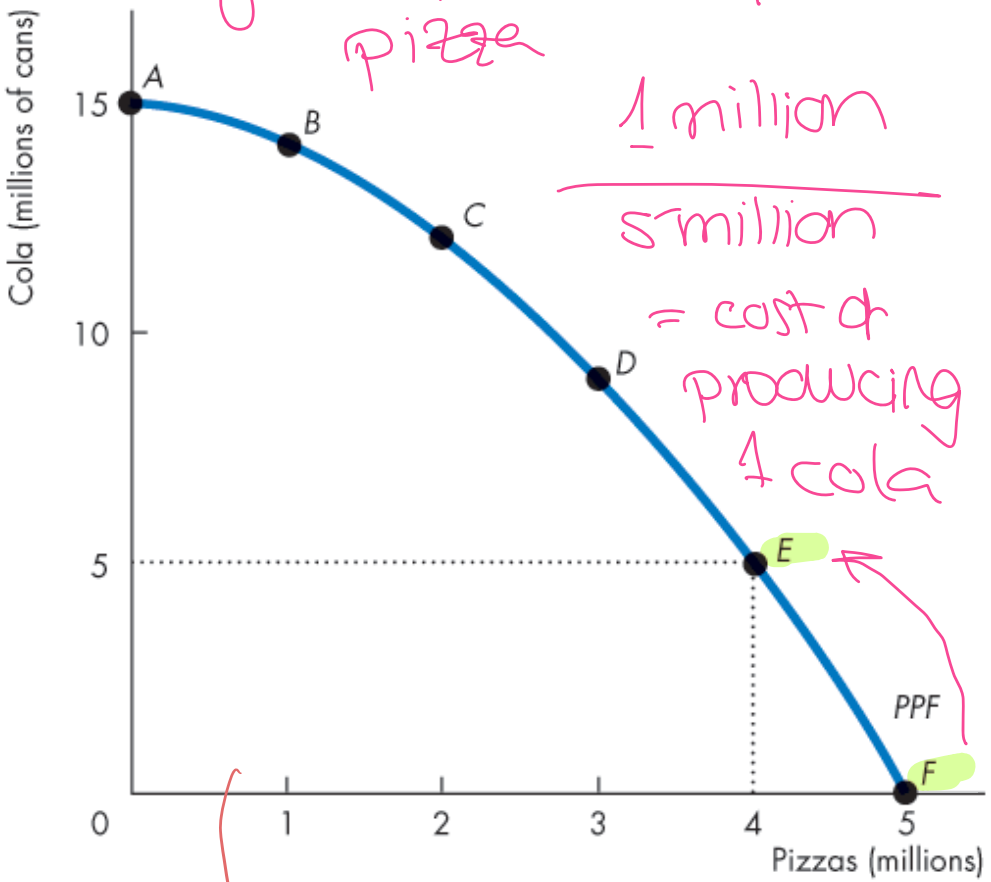
In moving from *F* to *E*:

The quantity of cola increases by 5 million cans.

The quantity of pizzas decreases by 1 million.

The opportunity cost of the first 5 million cans of cola is 1 million pizzas.

One of these cans of cola costs 1/5 of a pizza.



bow shaped due to increased opportunity costs

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Production Possibilities and Opportunity Cost

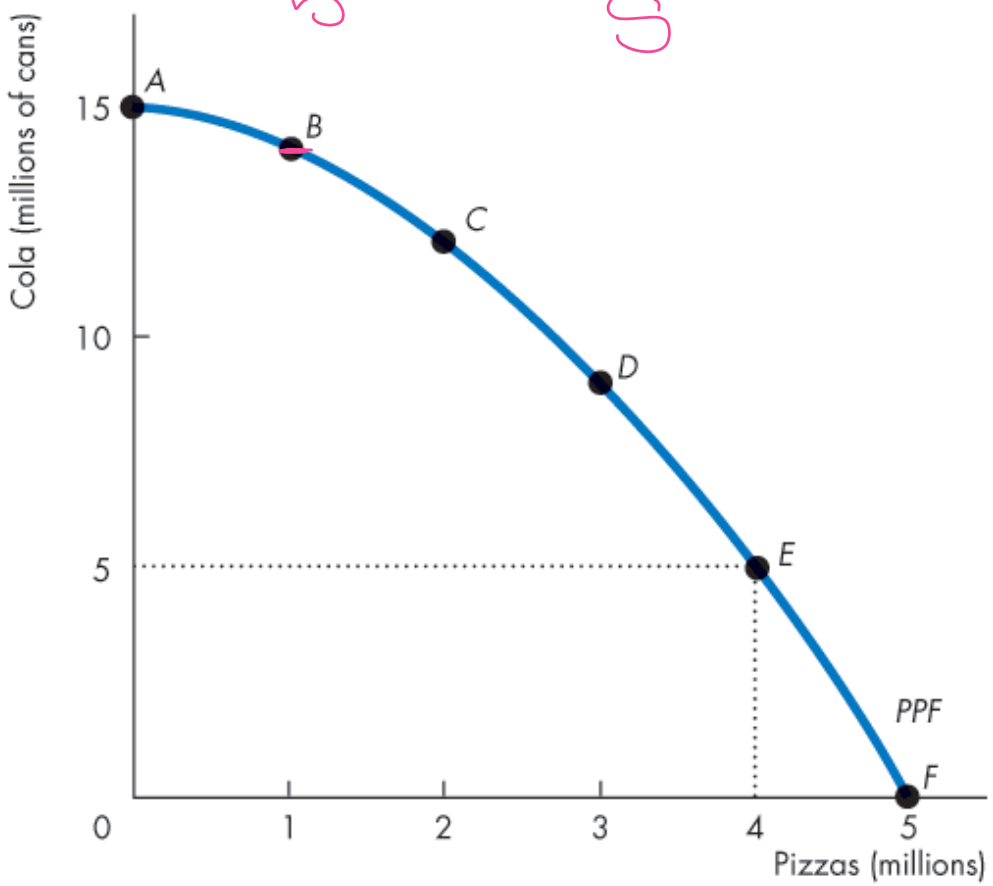
Opportunity Cost Is a Ratio

The opportunity cost of producing a can of cola is the *inverse* of the opportunity cost of producing a pizza.

One pizza costs 5 cans of cola.

One can of cola costs 1/5 of a pizza.

5 cans / 5 = 1 pizza / 5



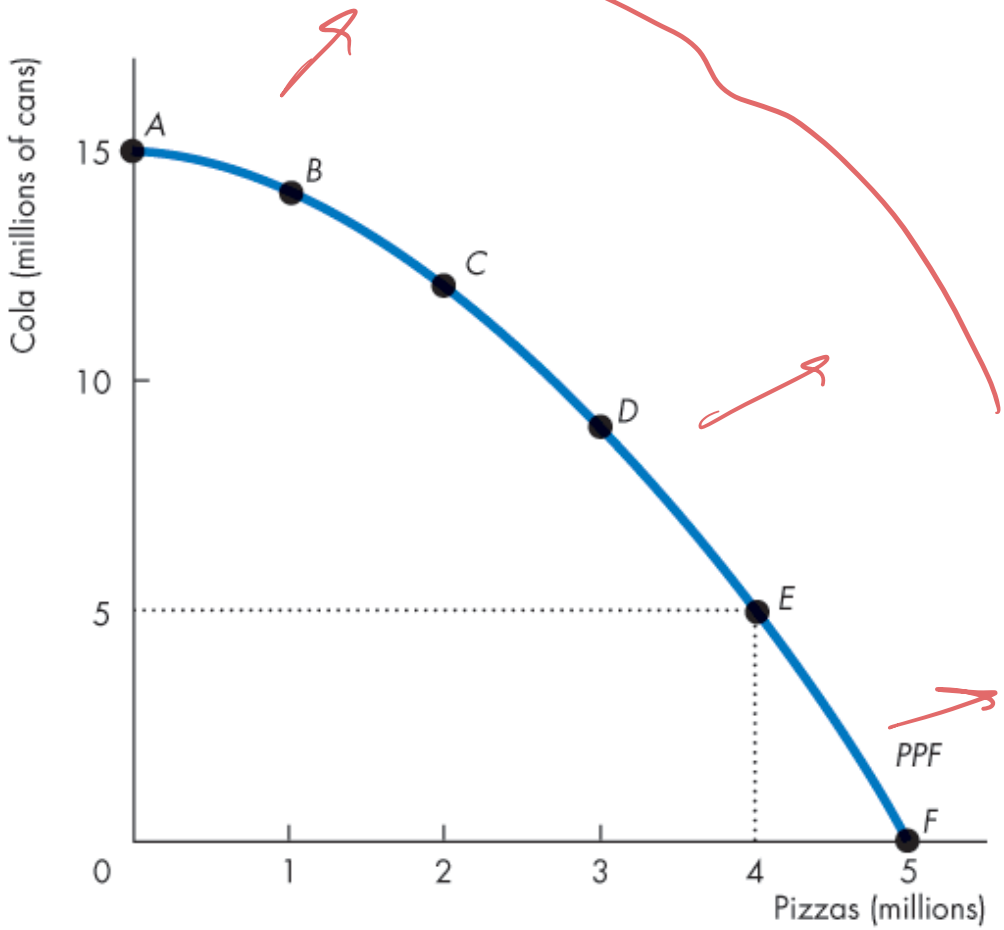
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Production Possibilities and Opportunity Cost

Increasing Opportunity Cost

Because resources are not equally productive in all activities, the *PPF* bows outward.

The outward bow of the *PPF* means that as the quantity produced of each good increases, so does its opportunity cost.



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Using Resources Efficiently

All the points along the *PPF* are efficient.

To determine which of the alternative efficient quantities to produce, we compare costs and benefits.

The *PPF* and Marginal Cost

The *PPF* determines opportunity cost.

The **marginal cost** of a good or service is the opportunity cost of producing *one more unit* of it.

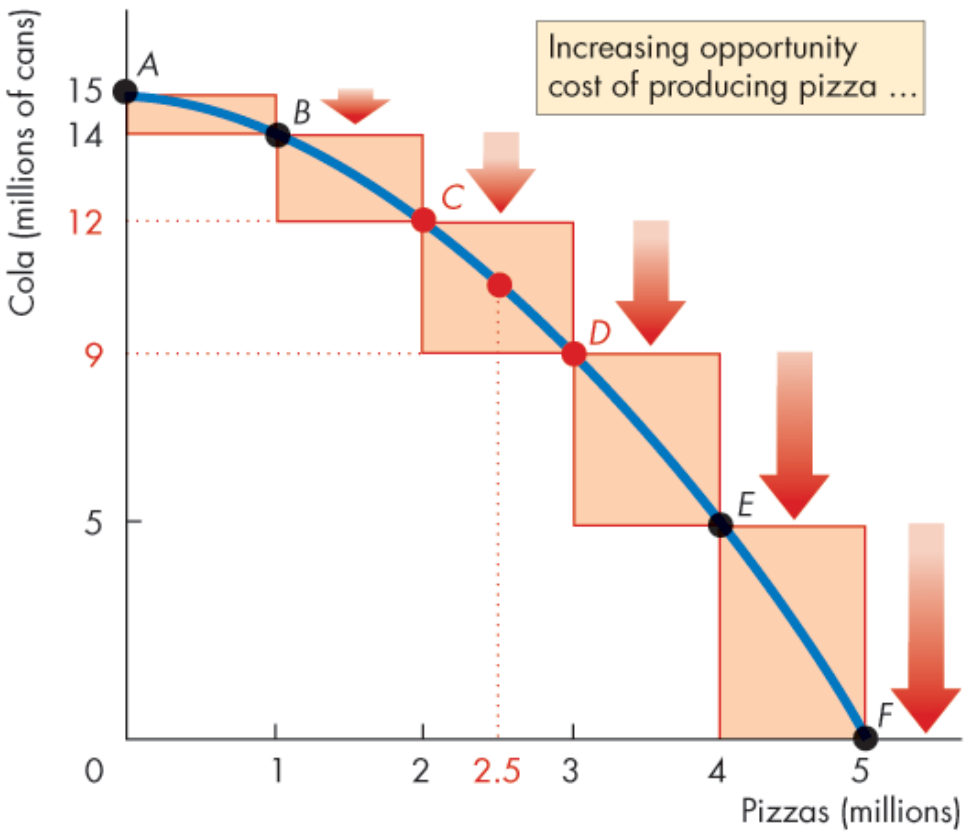
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Using Resources Efficiently

Figure 2.2 illustrates the marginal cost of a pizza.

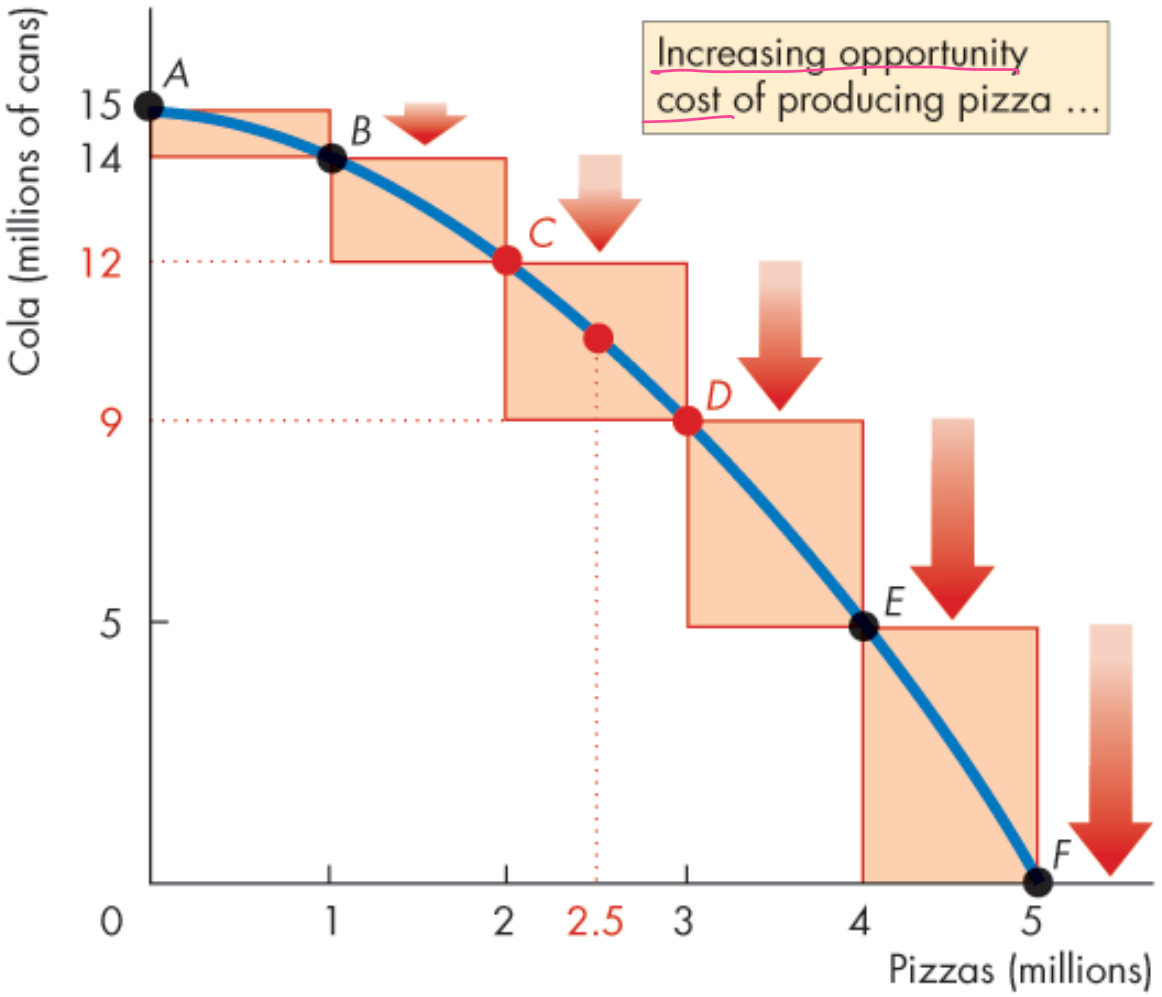
As we move along the *PPF*, the opportunity cost of a pizza increases.

The opportunity cost of producing one more pizza is the marginal cost of a pizza.



(a) *PPF* and opportunity cost

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(a) *PPF* and opportunity cost

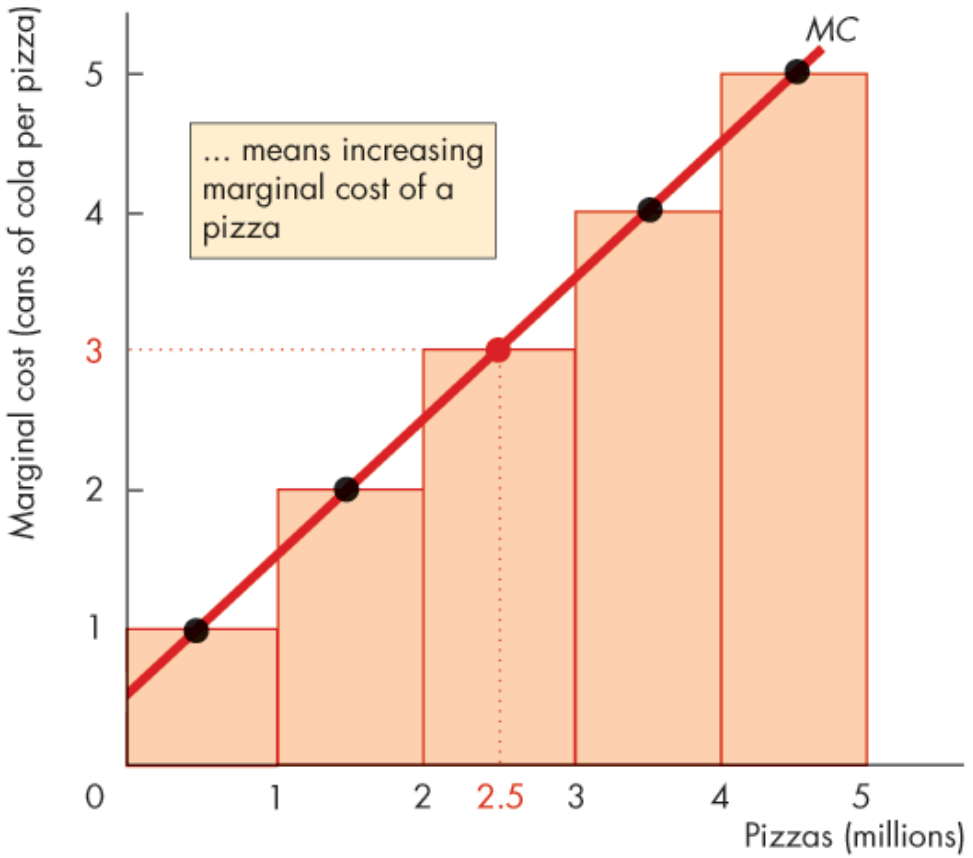
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Using Resources Efficiently

In part (b) of Fig. 2.2, the bars illustrate the increasing opportunity cost of a pizza.

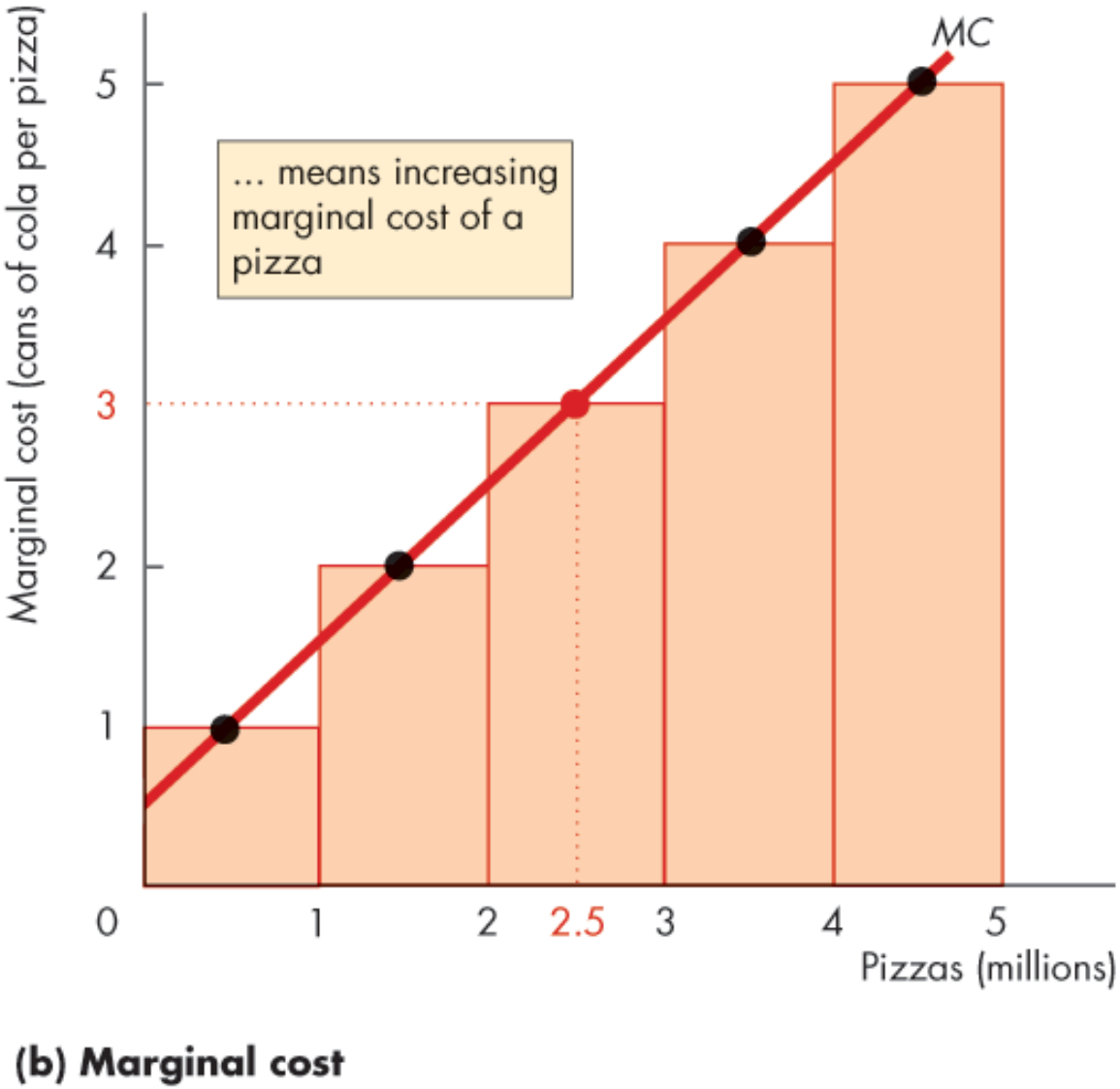
The black dots and the line *MC* show the marginal cost of producing a pizza.

The *MC* curve passes through the middle point of each bar.



(b) Marginal cost

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(b) Marginal cost

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Using Resources Efficiently

Preferences and Marginal Benefit

Preferences are a description of a person's likes and dislikes.

To describe preferences, economists use the concepts of marginal benefit and the marginal benefit curve.

The **marginal benefit** of a good or service is the benefit received from consuming one more unit of it.

We measure marginal benefit by the amount that a person is *willing to pay* for an additional unit of a good or service.

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Using Resources Efficiently

It is a general principle that:

The more we have of any good, the smaller is its marginal benefit and …

the less we are willing to pay for an additional unit of it.

We call this general principle the *principle of decreasing marginal benefit*.

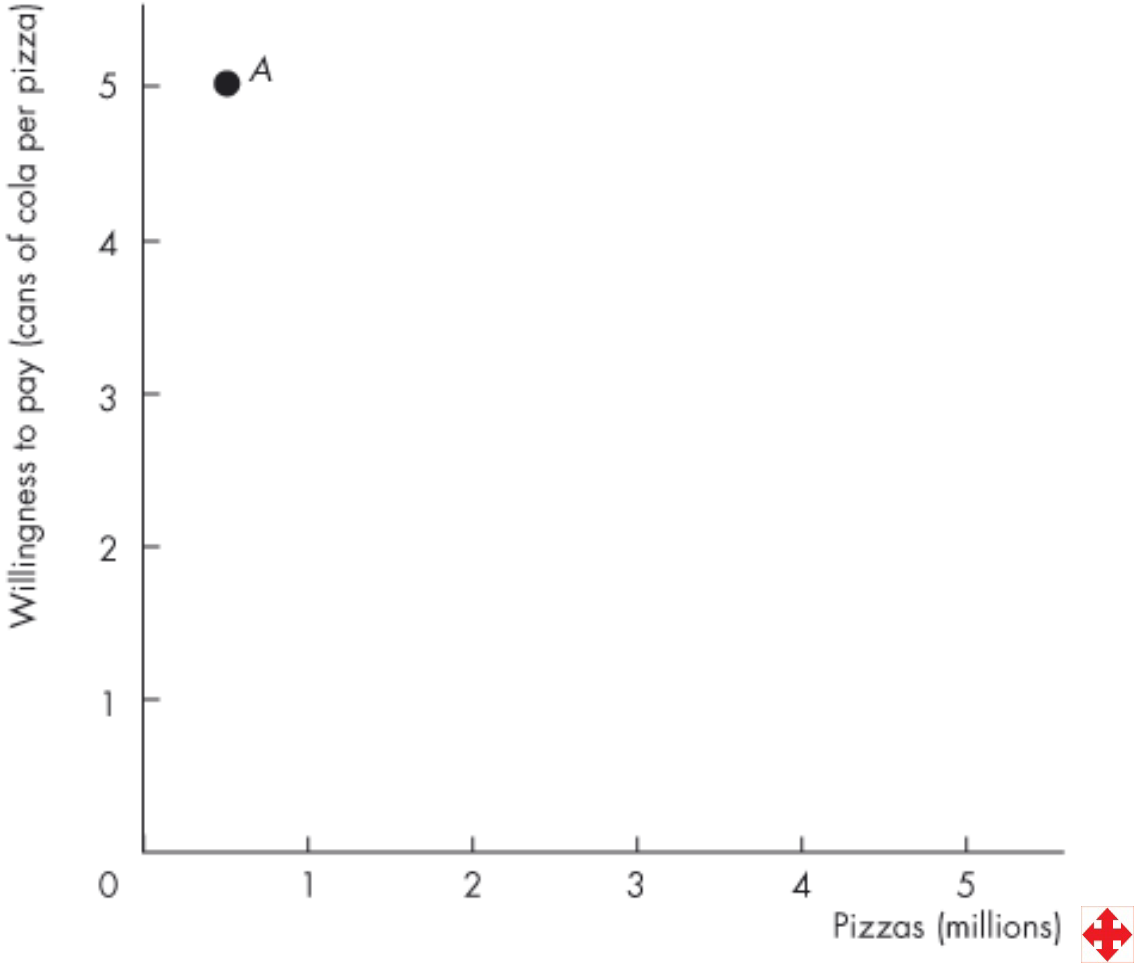
The **marginal benefit curve** shows the relationship between the marginal benefit of a good and the quantity of that good consumed.

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Using Resources Efficiently

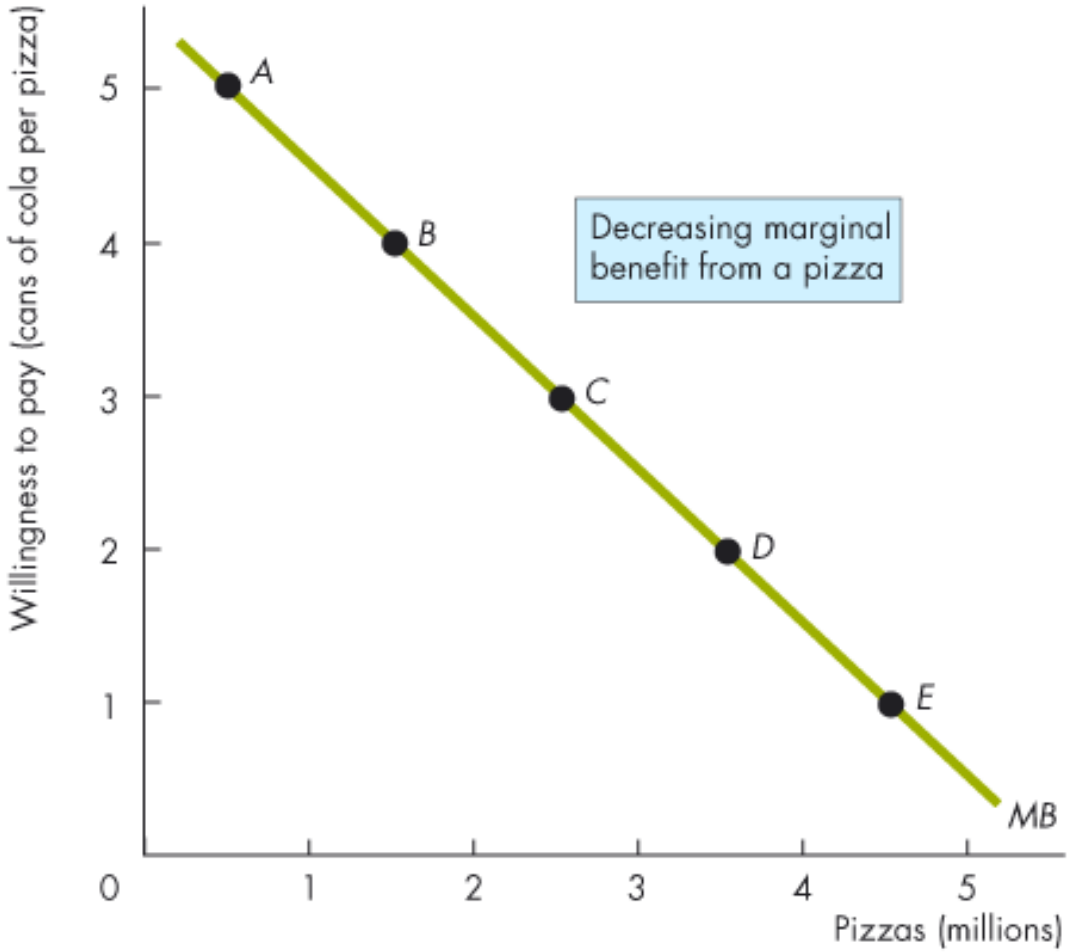
At point *A*, with 0.5 million pizzas available, people are willing to pay 5 cans of cola for a pizza.

Possibility	Pizzas (millions)	Willingness to pay (cans of cola per pizza)
A	0.5	5



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Possibility	Pizzas (millions)	Willingness to pay (cans of cola per pizza)
A	0.5	5
B	1.5	4
C	2.5	3
D	3.5	2
E	4.5	1



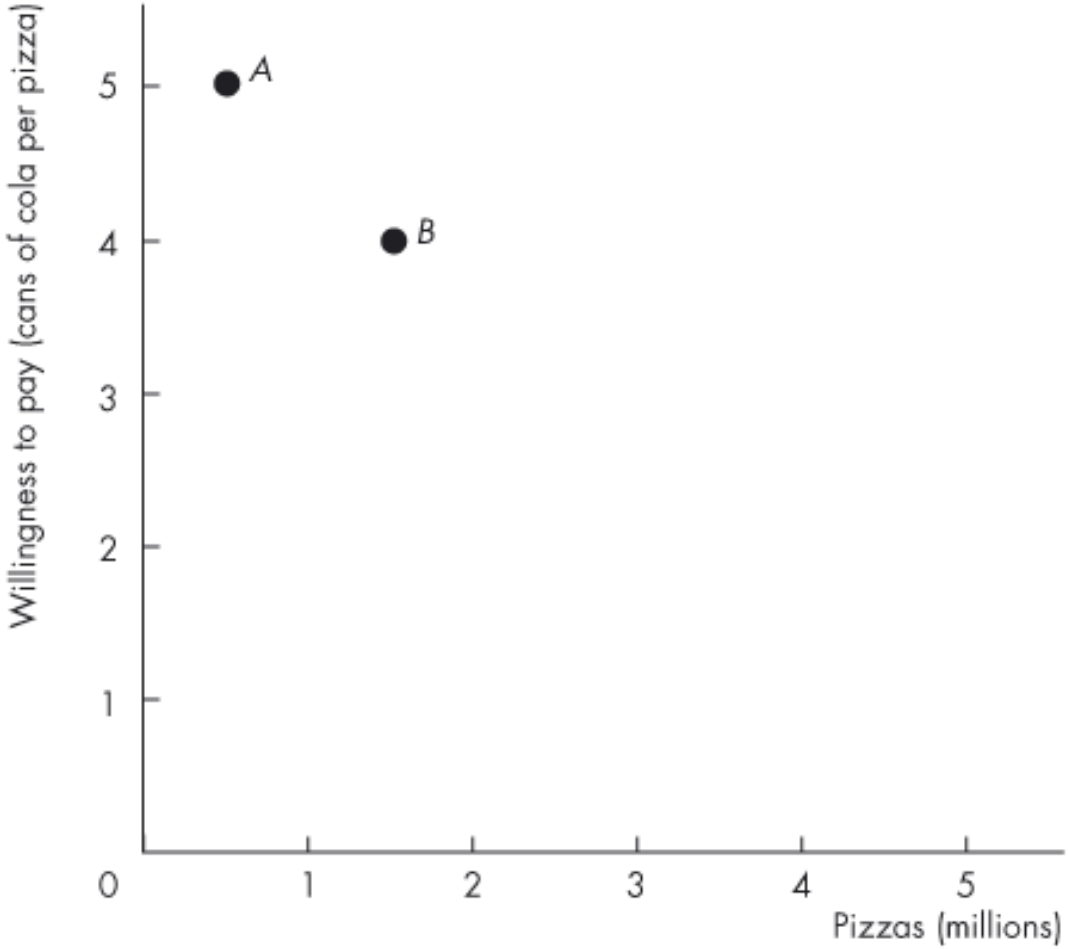
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Using Resources Efficiently

↳ should be equal to the marginal benefit of 1 item

At point *B*, with 1.5 million pizzas available, people are willing to pay 4 cans of cola for a pizza

Possibility	Pizzas (millions)	Willingness to pay (cans of cola per pizza)
A	0.5	5
B	1.5	4

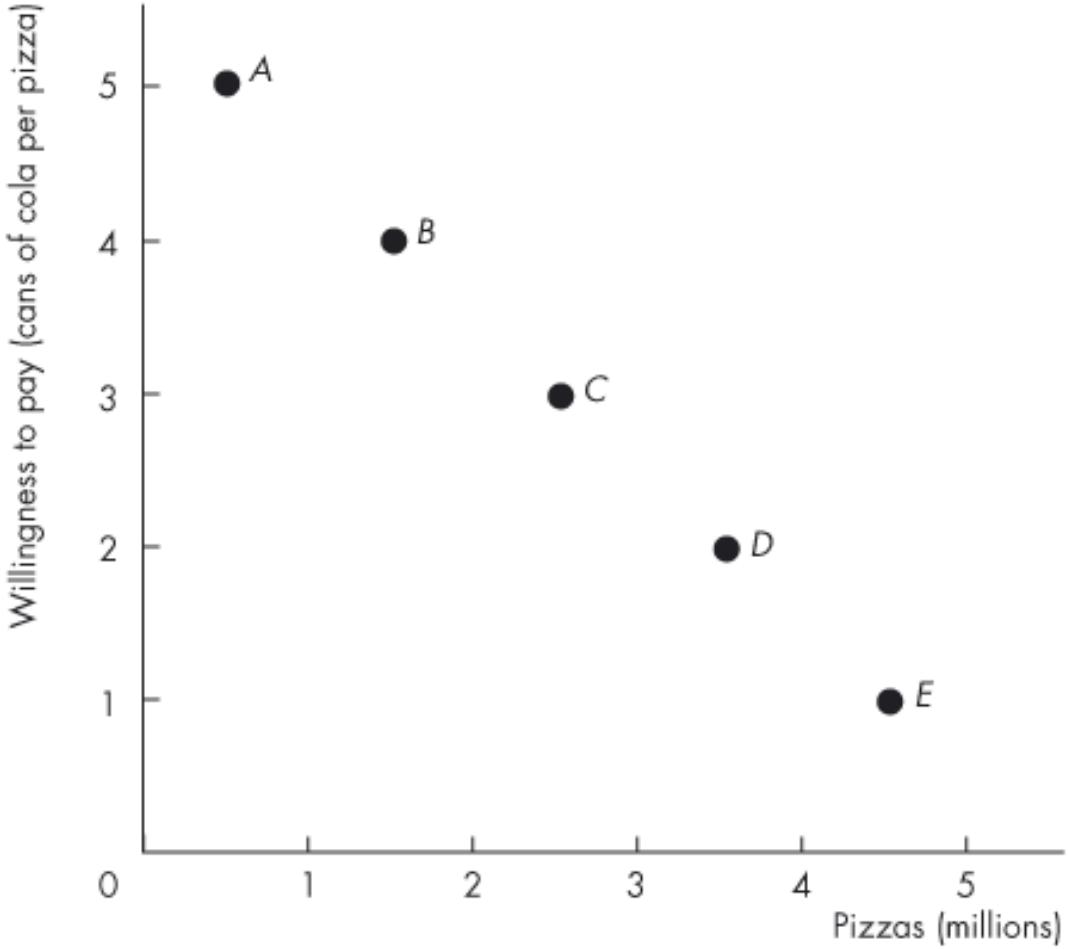


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Using Resources Efficiently

At point *E*, with 4.5 million pizzas available, people are willing to pay 1 can of cola for a pizza.

Possibility	Pizzas (millions)	Willingness to pay (cans of cola per pizza)
A	0.5	5
B	1.5	4
C	2.5	3
D	3.5	2
E	4.5	1

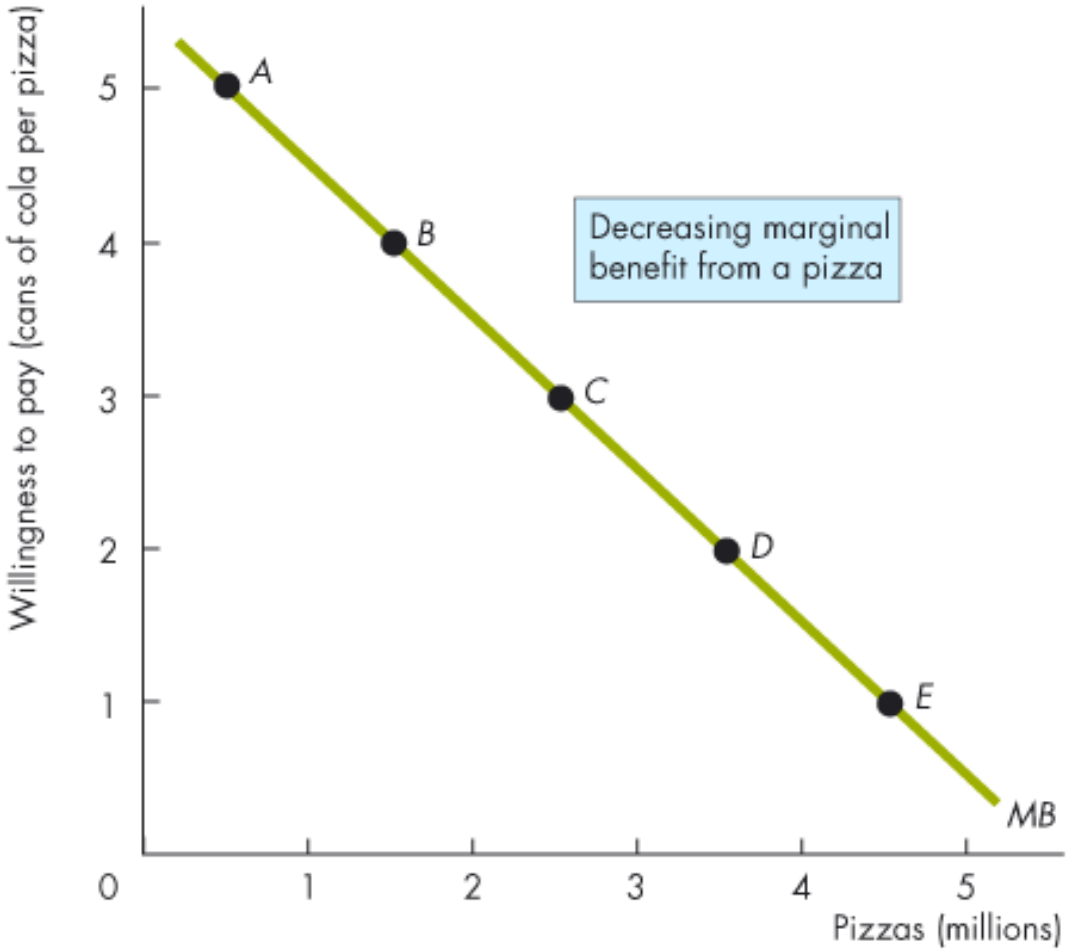


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Using Resources Efficiently

The line through the points shows the marginal benefit from a pizza.

Possibility	Pizzas (millions)	Willingness to pay (cans of cola per pizza)
A	0.5	5
B	1.5	4
C	2.5	3
D	3.5	2
E	4.5	1



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Using Resources Efficiently

Allocative Efficiency

When we cannot produce more of any one good without giving up some other good, we have achieved *production efficiency*.

We are producing at a point *on* the *PPF*.

When we cannot produce more of any one good without giving up some other good *that we value more highly*, we have achieved **allocative efficiency**.

We are producing at *the* point on the *PPF* that we prefer above all other points.

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Using Resources Efficiently

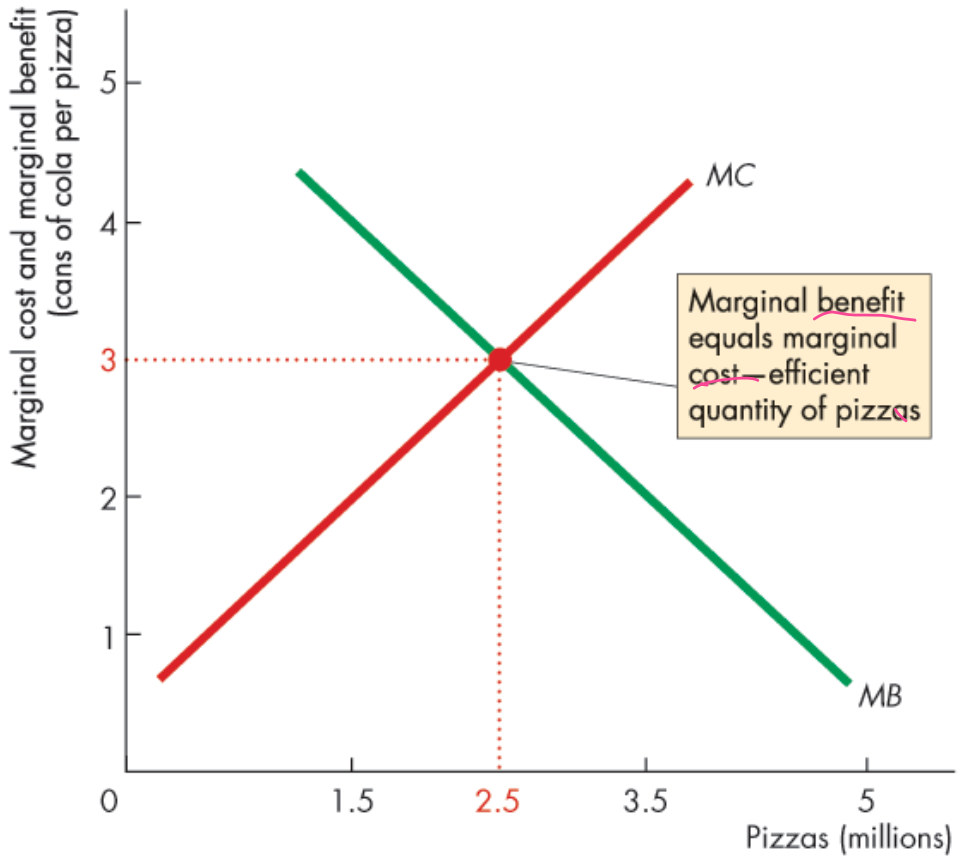
allocative efficiency occurs when $MB = MC$

Figure 2.4 illustrates allocative efficiency.

The point of allocative efficiency is the point on the *PPF* at which marginal benefit equals marginal cost.

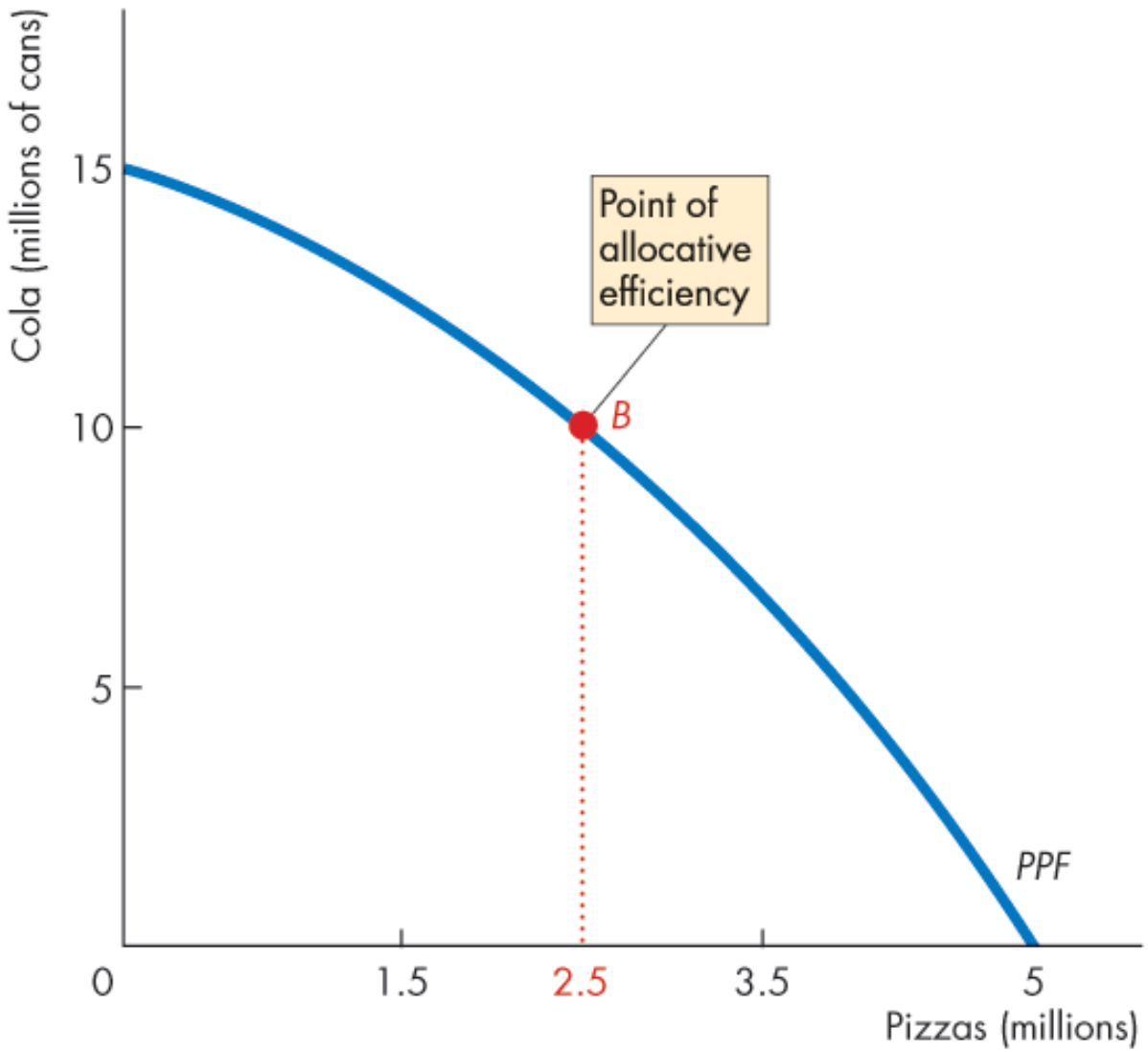
This point is determined by the quantity at which the marginal benefit curve intersects the marginal cost curve.

The efficient quantity is 2.5 million pizzas.



(b) Marginal benefit equals marginal cost

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(a) On the *PPF*

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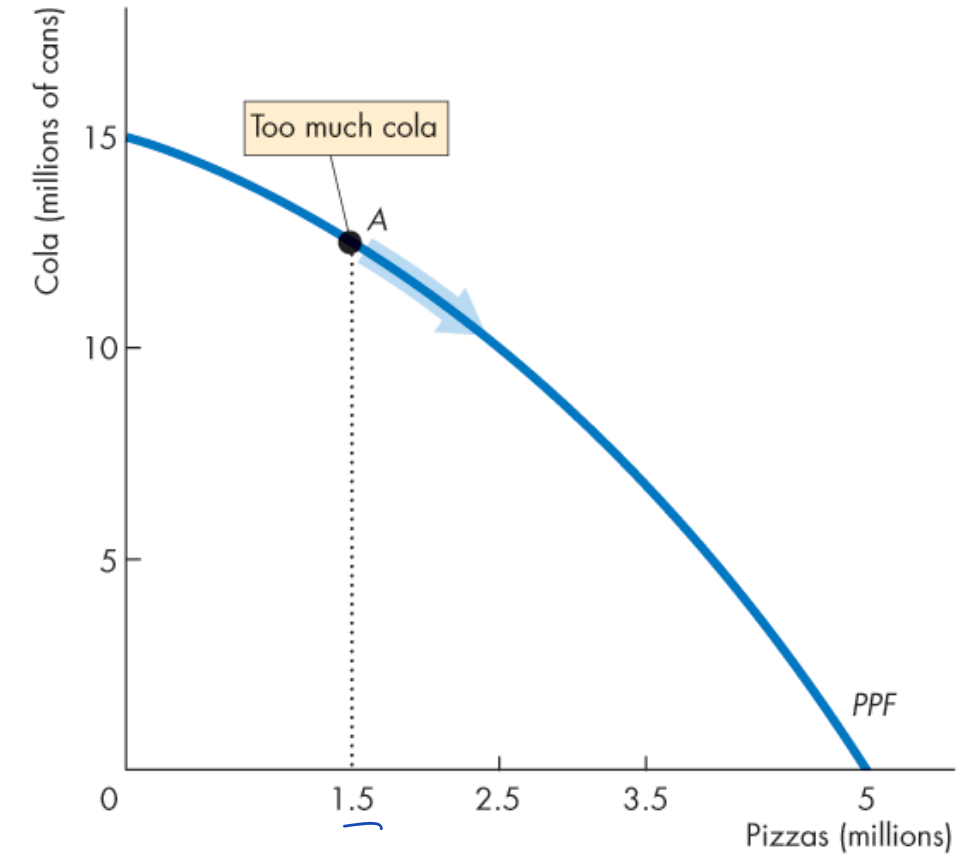
Using Resources Efficiently

If we produce 1.5 million pizzas, marginal benefit exceeds marginal cost.

We get more value from our resources by producing more pizzas.

On the *PPF* at point *A*, we produce too few pizzas.

We are better off moving along the *PPF* to produce more pizzas.



(a) On the *PPF*

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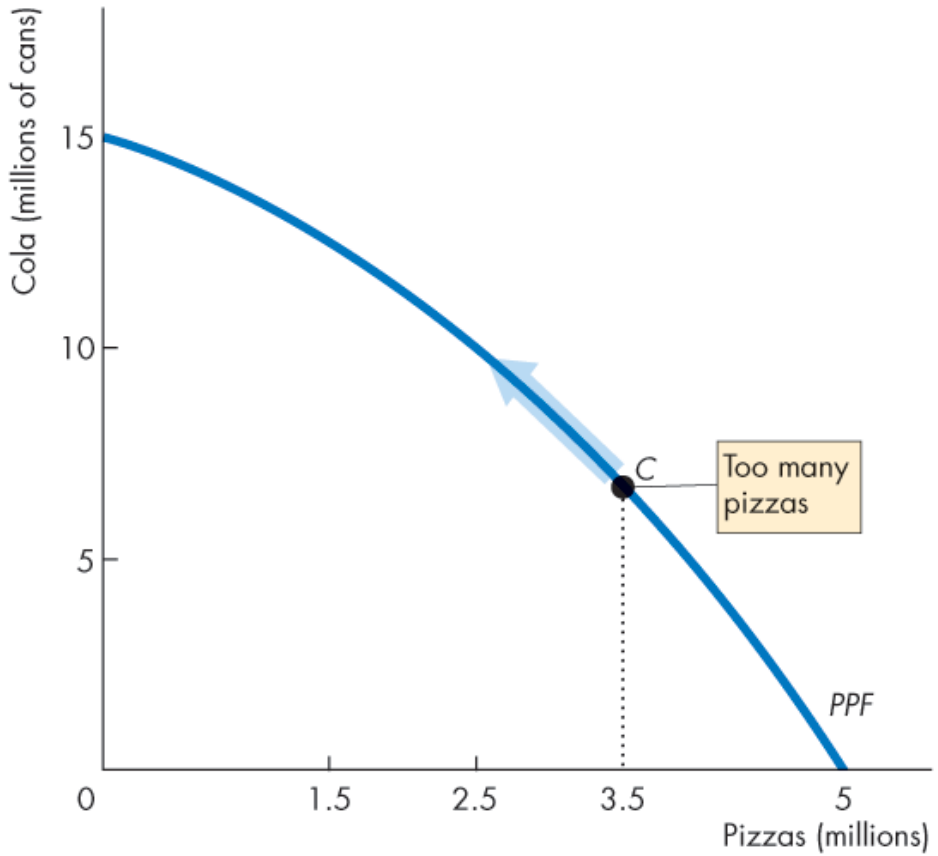
Using Resources Efficiently

If we produce 3.5 million pizzas, marginal cost exceeds marginal benefit.

We get more value from our resources by producing fewer pizzas.

On the *PPF* at point *C*, we produce too many pizzas.

We are better off moving along the *PPF* to produce fewer pizzas.



(a) On the *PPF*

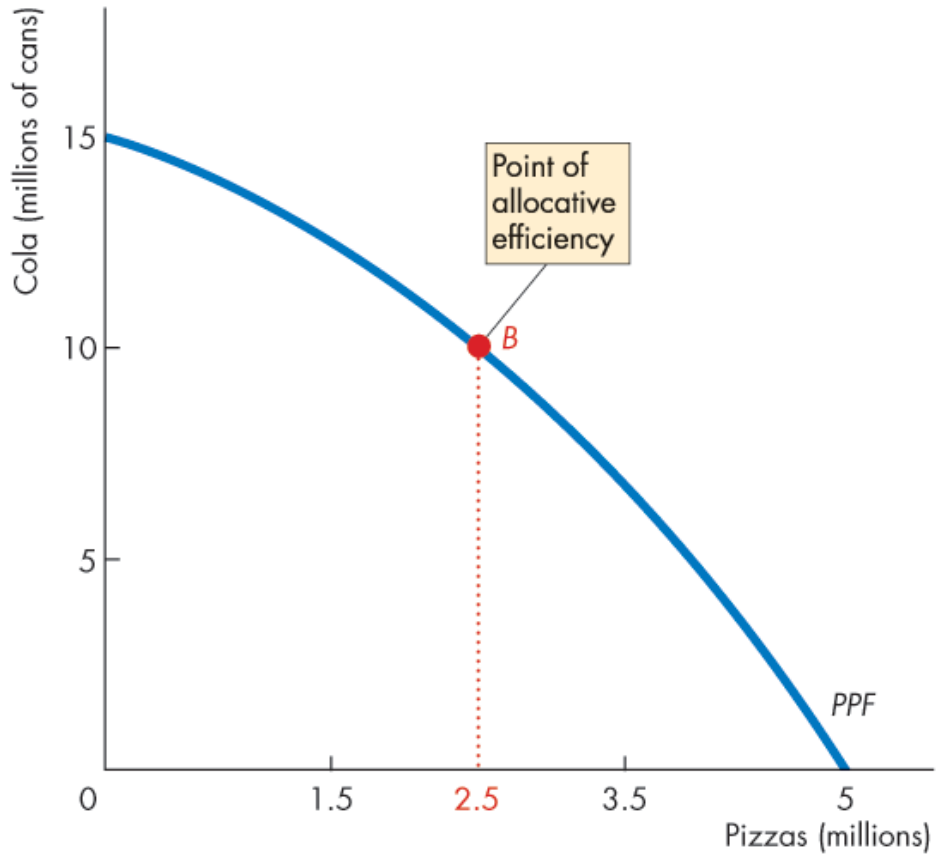
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Using Resources Efficiently

On the *PPF* at point *B*, we are producing the efficient quantities of pizzas and cola.

If we produce exactly 2.5 million pizzas, marginal cost equals marginal benefit.

We cannot get more value from our resources.



(a) On the *PPF*

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Gains from Trade → exports / imports

Comparative Advantage and Absolute Advantage

↑ comparing an individuals production with a low opportunity cost

A person has a **comparative advantage** in an activity if that person can perform the activity at a lower opportunity cost than anyone else. → you don't have to sacrifice as much

A person has an **absolute advantage** if that person is more productive than others. → those who can produce more

↳ meaning that may tend to produce more than others

Absolute advantage involves comparing productivities while comparative advantage involves comparing opportunity costs.

Let's look at Joe and Liz who operate smoothie bars.

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Gains from Trade

Joe's Smoothie Bar

In an hour, Joe can produce 6 smoothies or 30 salads.

Joe's opportunity cost of producing 1 smoothie is 5 salads.

TABLE 2.1 Joe's Production Possibilities

Item	Minutes to produce 1	Quantity per hour
Smoothies	10	6
Salads	2	30

Joe's opportunity cost of producing 1 salad is 1/5 smoothie.

Joe spends 10 minutes making salads and 50 minutes making smoothies, so he produces 5 smoothies and 5 salads an hour.

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Gains from Trade

Liz's Smoothie Bar

In an hour, Liz can produce 30 smoothies or 30 salads.

Liz's opportunity cost of producing 1 smoothie is 1 salad.

TABLE 2.2 Liz's Production Possibilities

Item	Minutes to produce 1 item	Quantity per hour
Smoothies	2	30
Salads	2	30

Liz's opportunity cost of producing 1 salad is 1 smoothie.

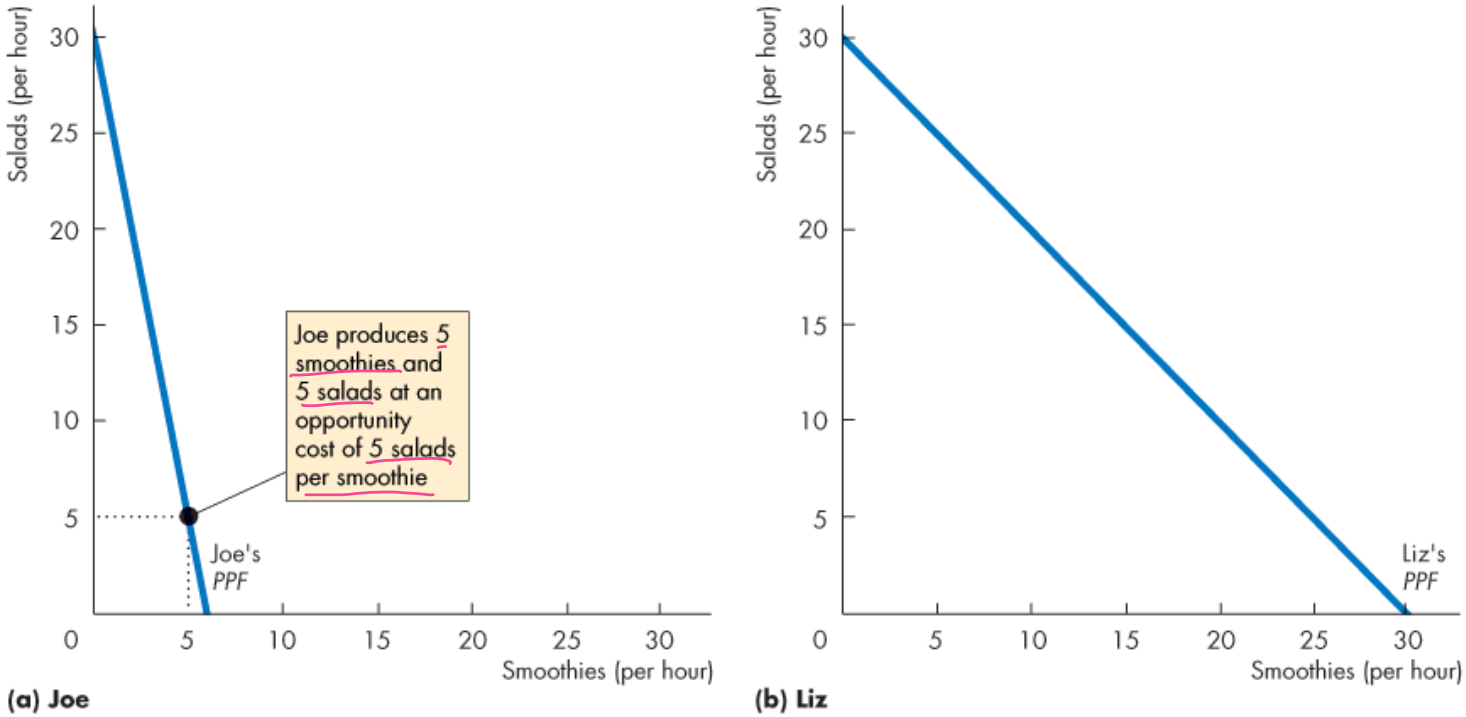
Liz's customers buy salads and smoothies in equal number, so she produces 15 smoothies and 15 salads an hour.

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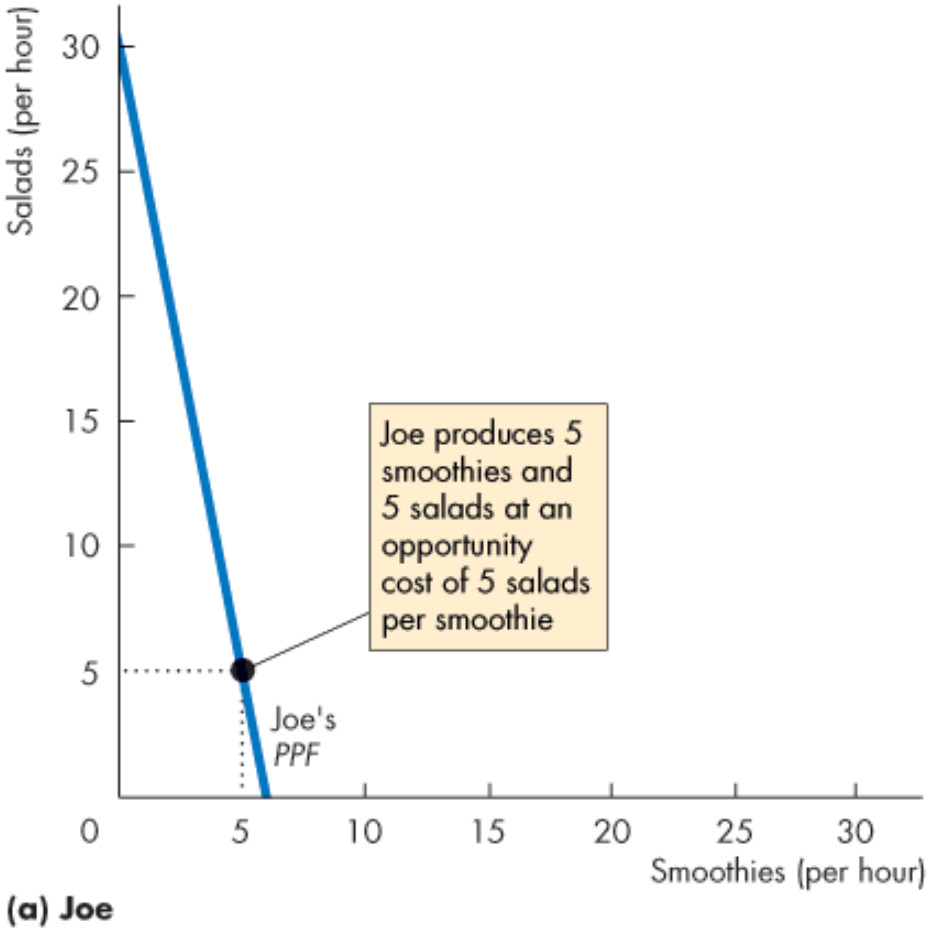
Gains from Trade

Figure 2.5 shows the production possibility frontiers.

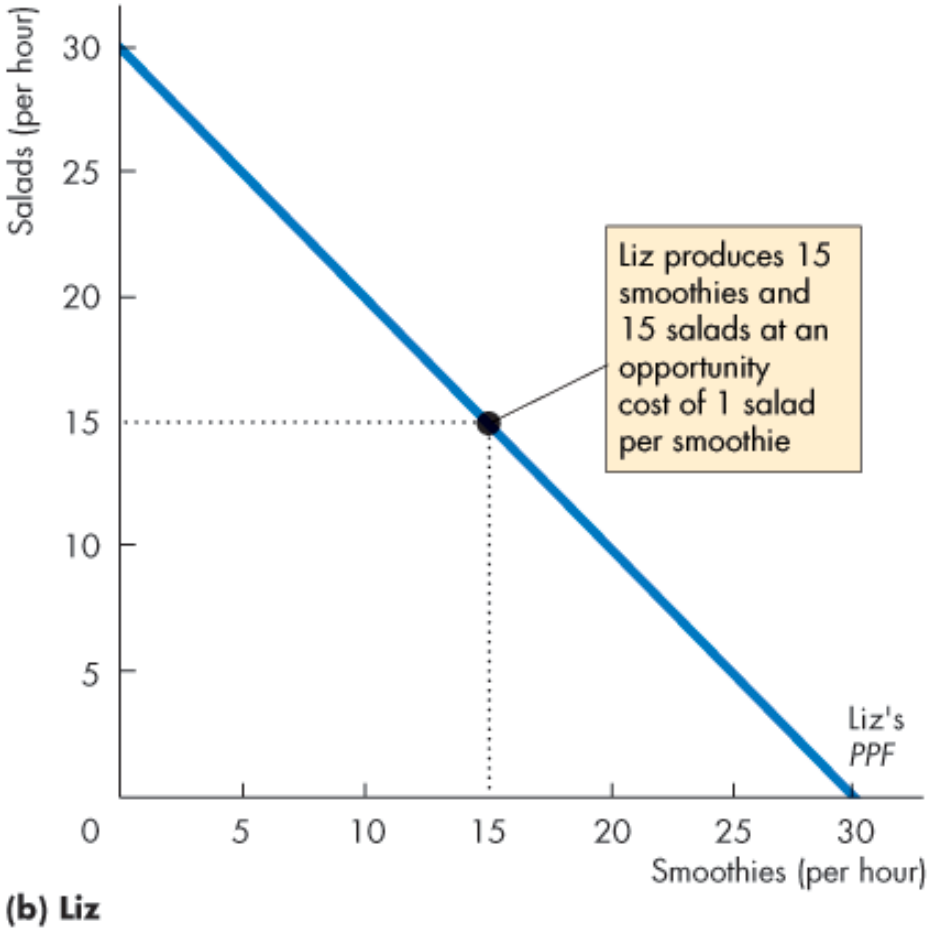
In part (a), Joe's opportunity cost of a smoothie is 5 salads. Joe produces at a point on his *PPF.*



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(a) Joe



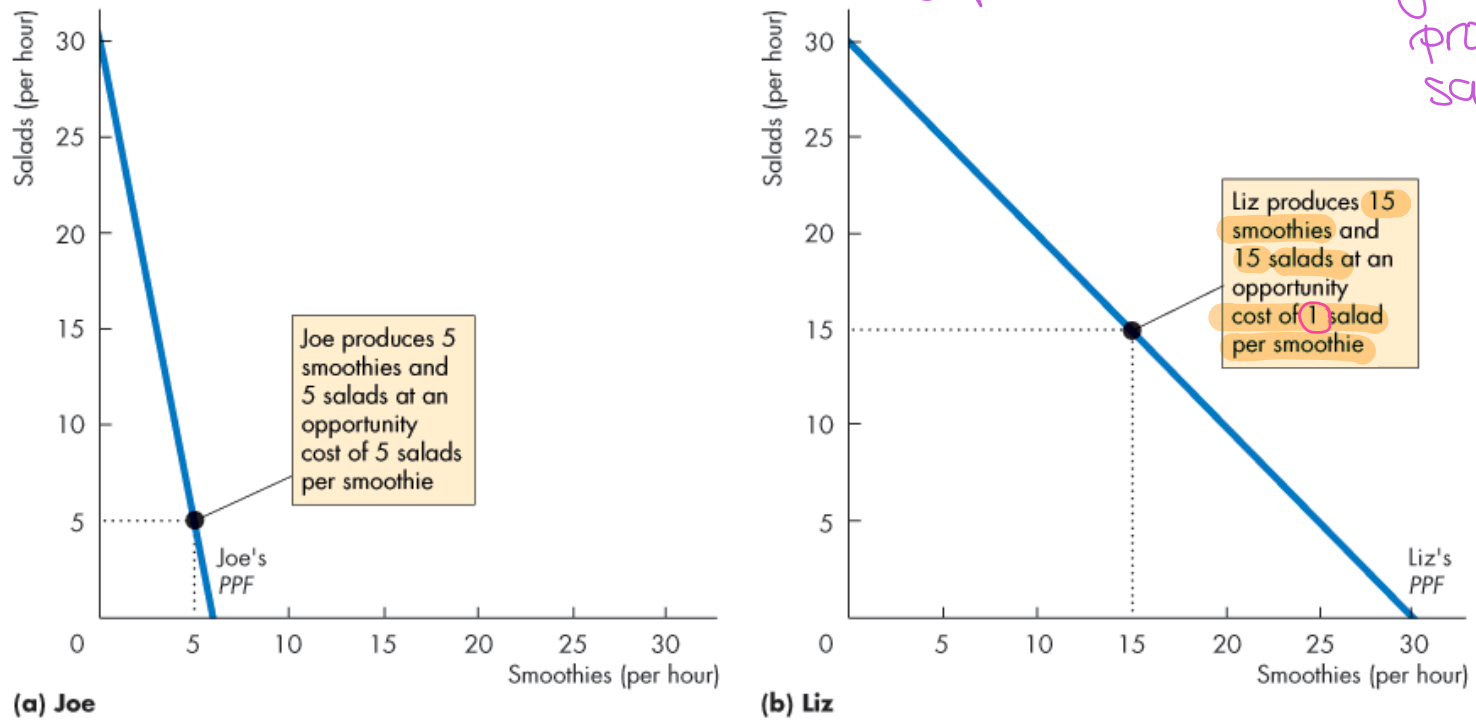
(b) Liz

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Gains from Trade

In part (b), Liz's opportunity cost of a smoothie is 1 salad. Liz produces at a point on her *PPF.*

Joe: 1 smoothie → 5 salads
Liz: 1 smoothie → 1 salad
↓ she has the comparative advantage of producing smoothie while Joe has more comparative advantage of producing salads



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Gains from Trade

Joe's Comparative Advantage

Joe's opportunity cost of a salad is 1/5 smoothie.

Liz's opportunity cost of a salad is 1 smoothie.

Joe's opportunity cost of a salad is less than Liz's.

So Joe has a comparative advantage in producing salads.

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Gains from Trade

Liz's Comparative Advantage

Liz's opportunity cost of a smoothie is 1 salad.

Joe's opportunity cost of a smoothie is 5 salads.

Liz's opportunity cost of a smoothie is less than Joe's.

So Liz has a comparative advantage in producing smoothies.

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Gains from Trade

Achieving the Gains from Trade

Liz and Joe produce the good in which they have a comparative advantage:

- Liz produces 30 smoothies and 0 salads.
- Joe produces 30 salads and 0 smoothies.

TABLE 2.3 Liz and Joe Gain from Trade

(a) Before trade	Liz	Joe
Smoothies	15	5
Salads	15	5
(b) Specialization	**Liz**	**Joe**
Smoothies	30	0
Salads	0	30

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TABLE 2.3 Liz and Joe Gain from Trade

(a) Before trade	Liz	Joe
Smoothies	15	5
Salads	15	5

(b) Specialization	Liz	Joe
Smoothies	30 - 10 = 20	0 + 10 = 10
Salads	0 + 20 = 20	30 - 20 = 10

(c) Trade	Liz	Joe
Smoothies	sell 10	buy 10
Salads	buy 20	sell 20

(d) After trade	Liz	Joe
Smoothies	20 → +5	10 → +5
Salads	20 → +5	10 → +5

(e) Gains from trade	Liz	Joe
Smoothies	+5	+5
Salads	+5	+5

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Gains from Trade

Liz and Joe trade:

- Liz sells Joe 10 smoothies and buys 20 salads.
- Joe sells Liz 20 salads and buys 10 smoothies.

After trade:

- Liz has 20 smoothies and 20 salads.
- Joe has 10 smoothies and 10 salads.

TABLE 2.3 Liz and Joe Gain from Trade

(a) Before trade	Liz	Joe
Smoothies	15	5
Salads	15	5
(b) Specialization	**Liz**	**Joe**
Smoothies	30	0
Salads	0	30
(c) Trade	**Liz**	**Joe**
Smoothies	sell 10	buy 10
Salads	buy 20	sell 20
(d) After trade	**Liz**	**Joe**
Smoothies	20	10
Salads	20	10

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Gains from Trade

Gains from trade:

- Liz gains 5 smoothies and 5 salads an hour
- Joe gains 5 smoothies and 5 salads an hour

TABLE 2.3 Liz and Joe Gain from Trade

(a) Before trade	Liz	Joe
Smoothies	15	5
Salads	15	5
(b) Specialization	**Liz**	**Joe**
Smoothies	30	0
Salads	0	30
(c) Trade	**Liz**	**Joe**
Smoothies	sell 10	buy 10
Salads	buy 20	sell 20
(d) After trade	**Liz**	**Joe**
Smoothies	20	10
Salads	20	10
(e) Gains from trade	**Liz**	**Joe**
Smoothies	+5	+5
Salads	+5	+5

more imp tends to reduce GDP

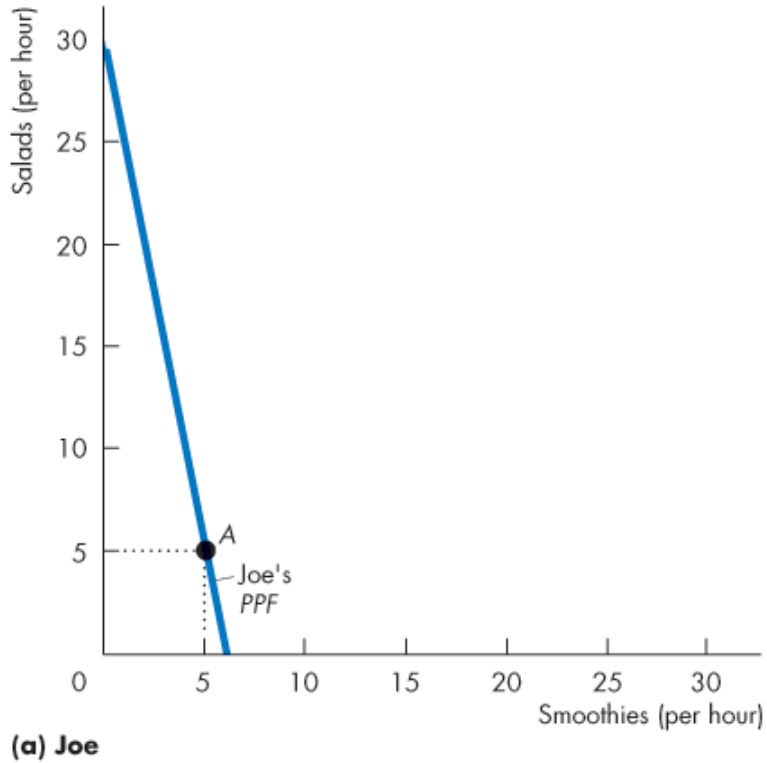
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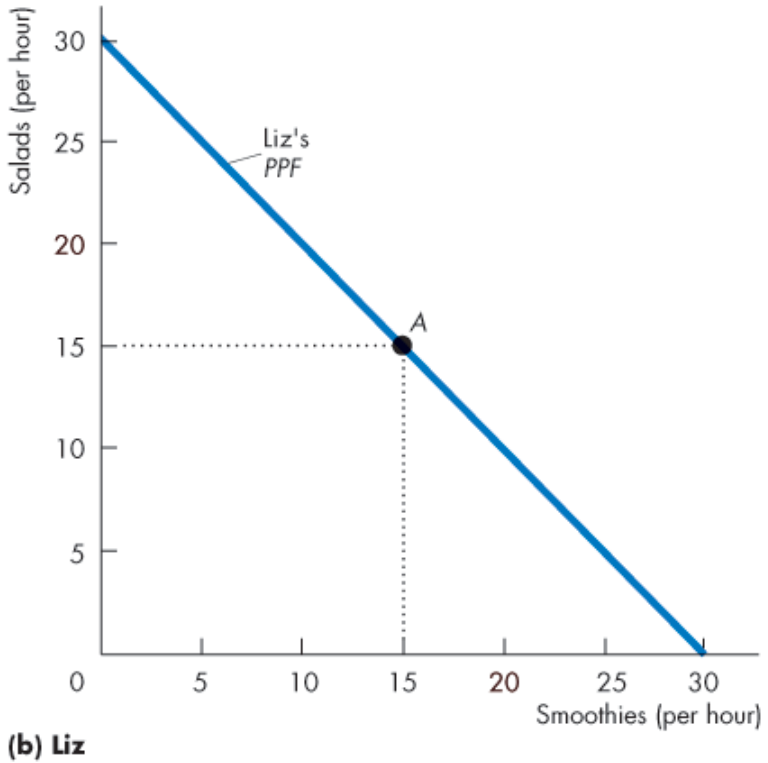
Gains from Trade

Figure 2.6 shows the gains from trade.

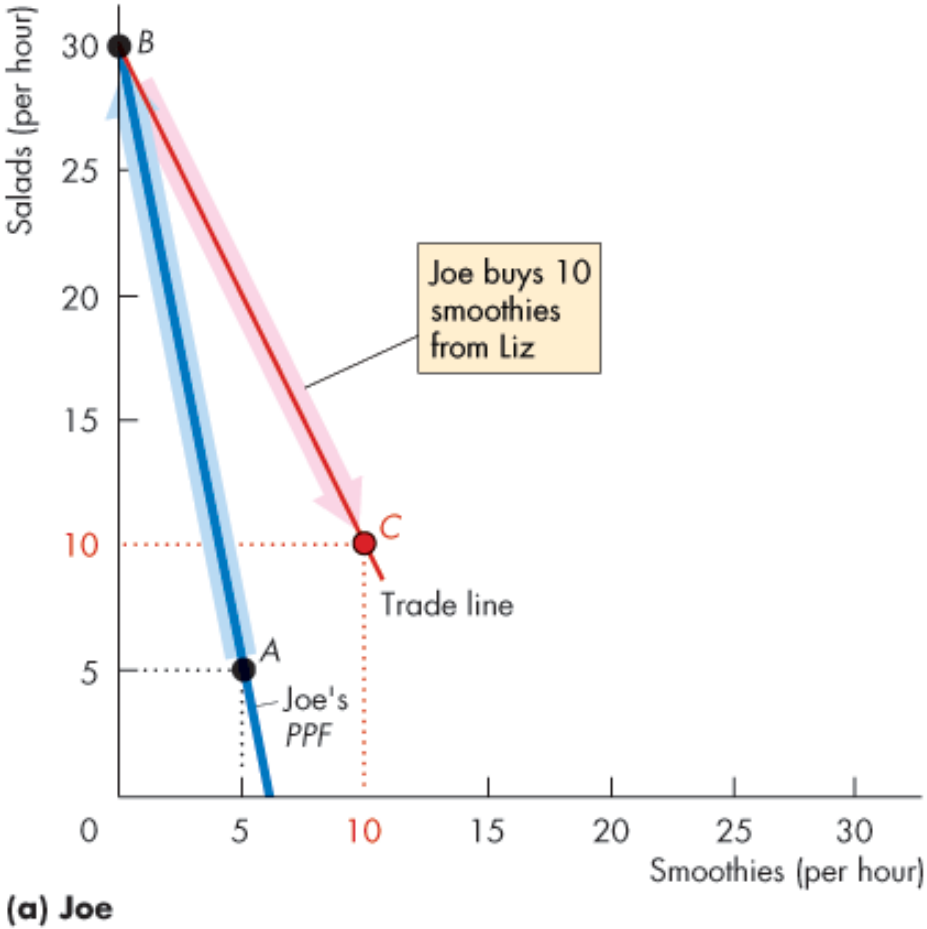
Joe's opportunity cost of producing a salad is less than Liz's.

So Joe has a comparative advantage in producing salad.

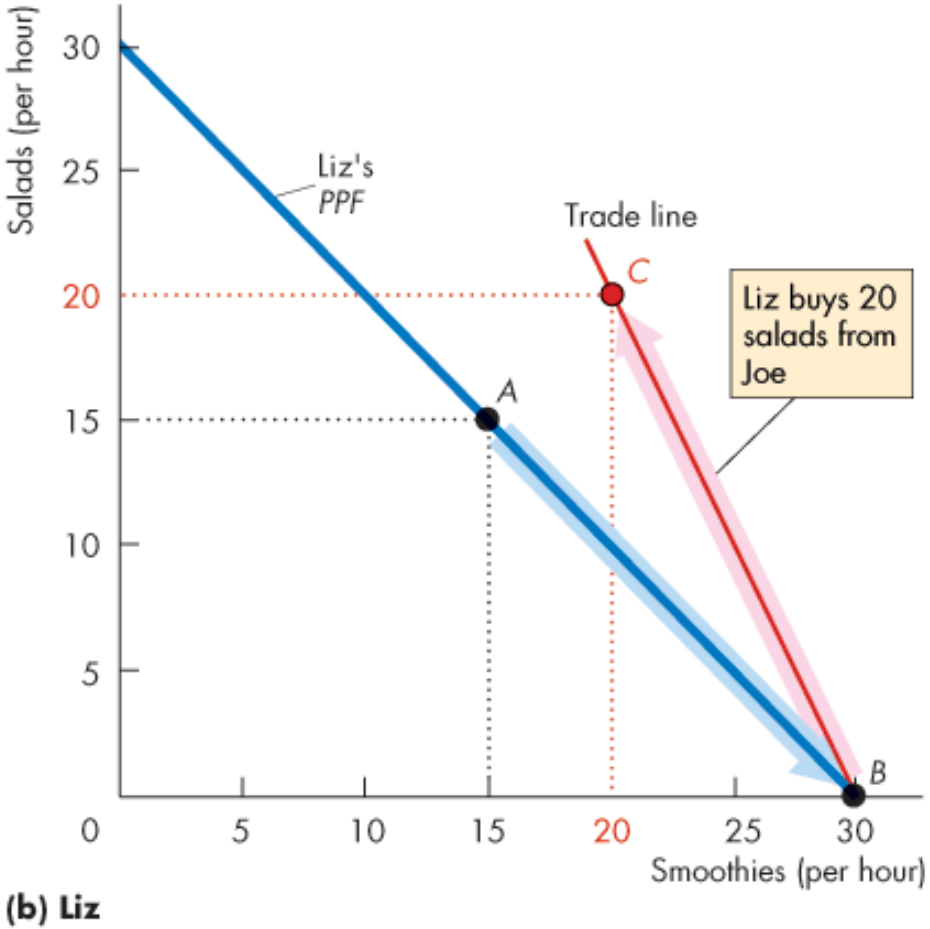




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(a) Joe



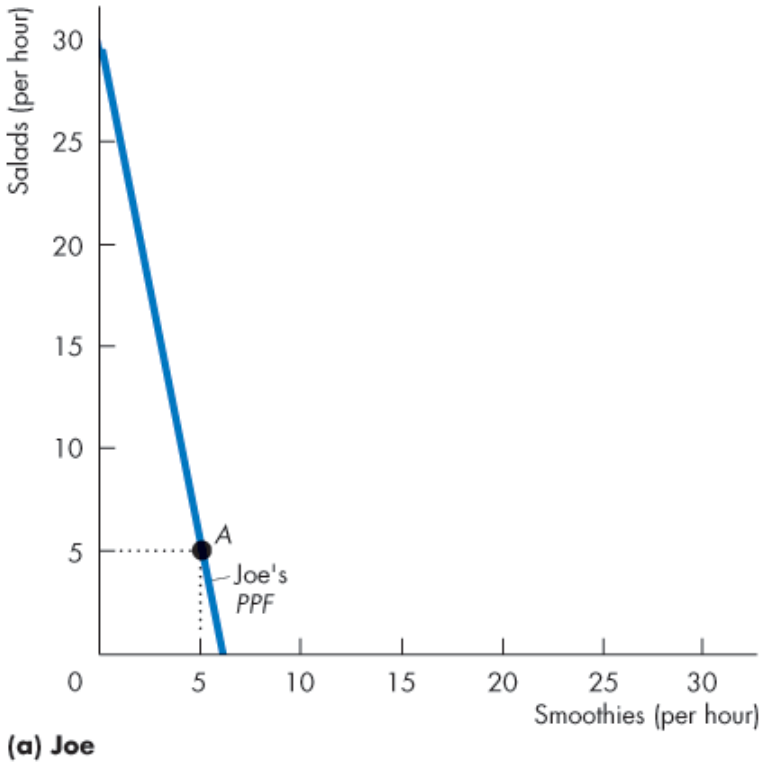
(b) Liz

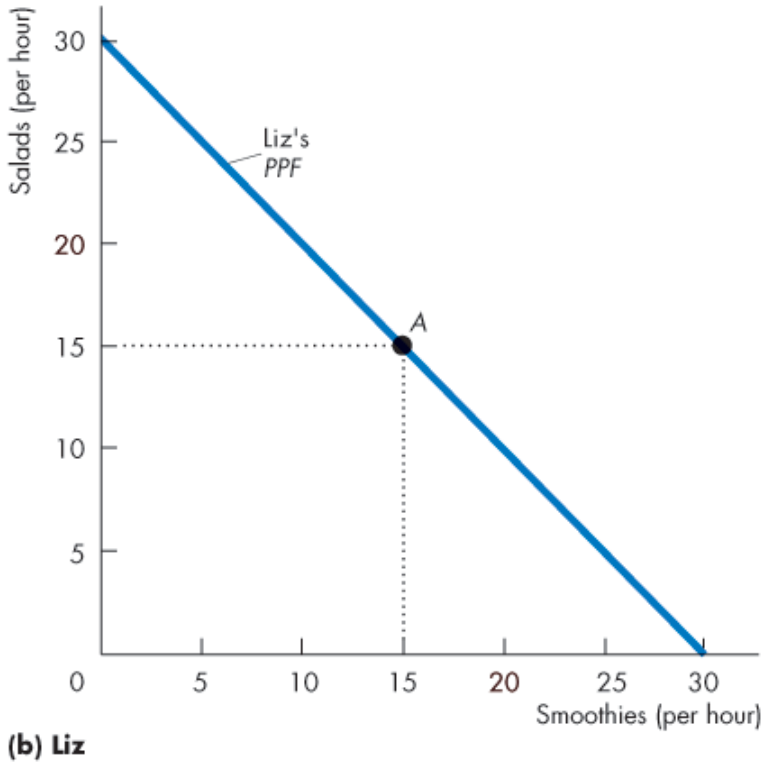
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Gains from Trade

Liz's opportunity cost of producing a smoothie is less than Joe's.

So Liz has a comparative advantage in producing smoothies.



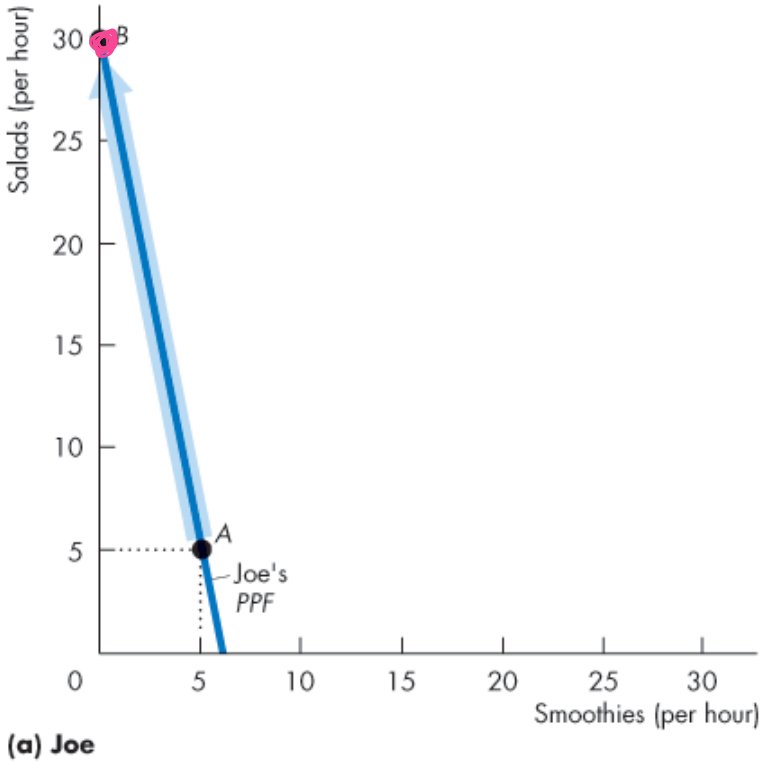


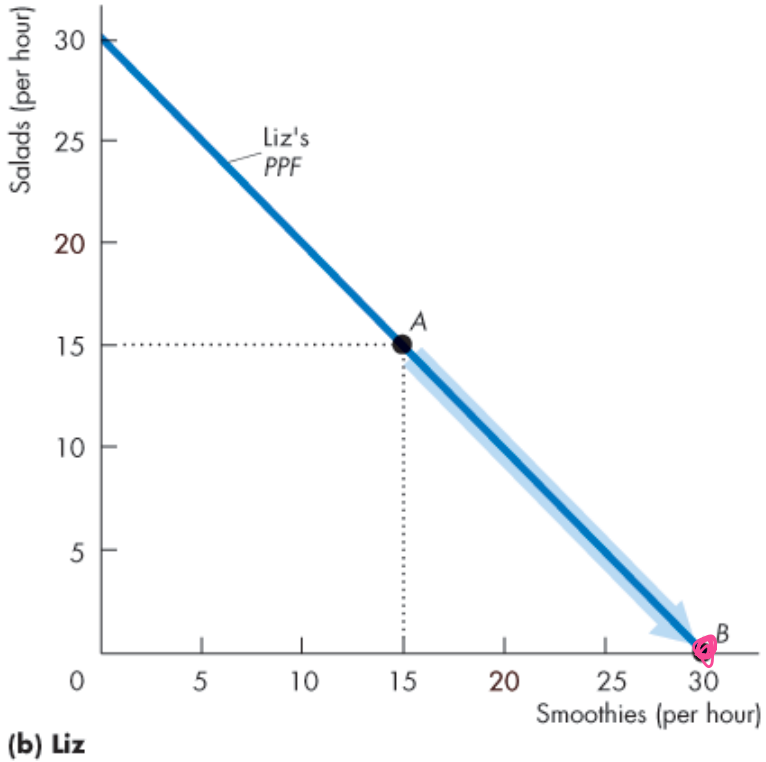
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Gains from Trade

Joe specializes in producing salad and produces 30 salads an hour at point *B* on his *PPF*.

Liz specializes in producing smoothies and produces 30 smoothies an hour at point *B* on her *PPF*.



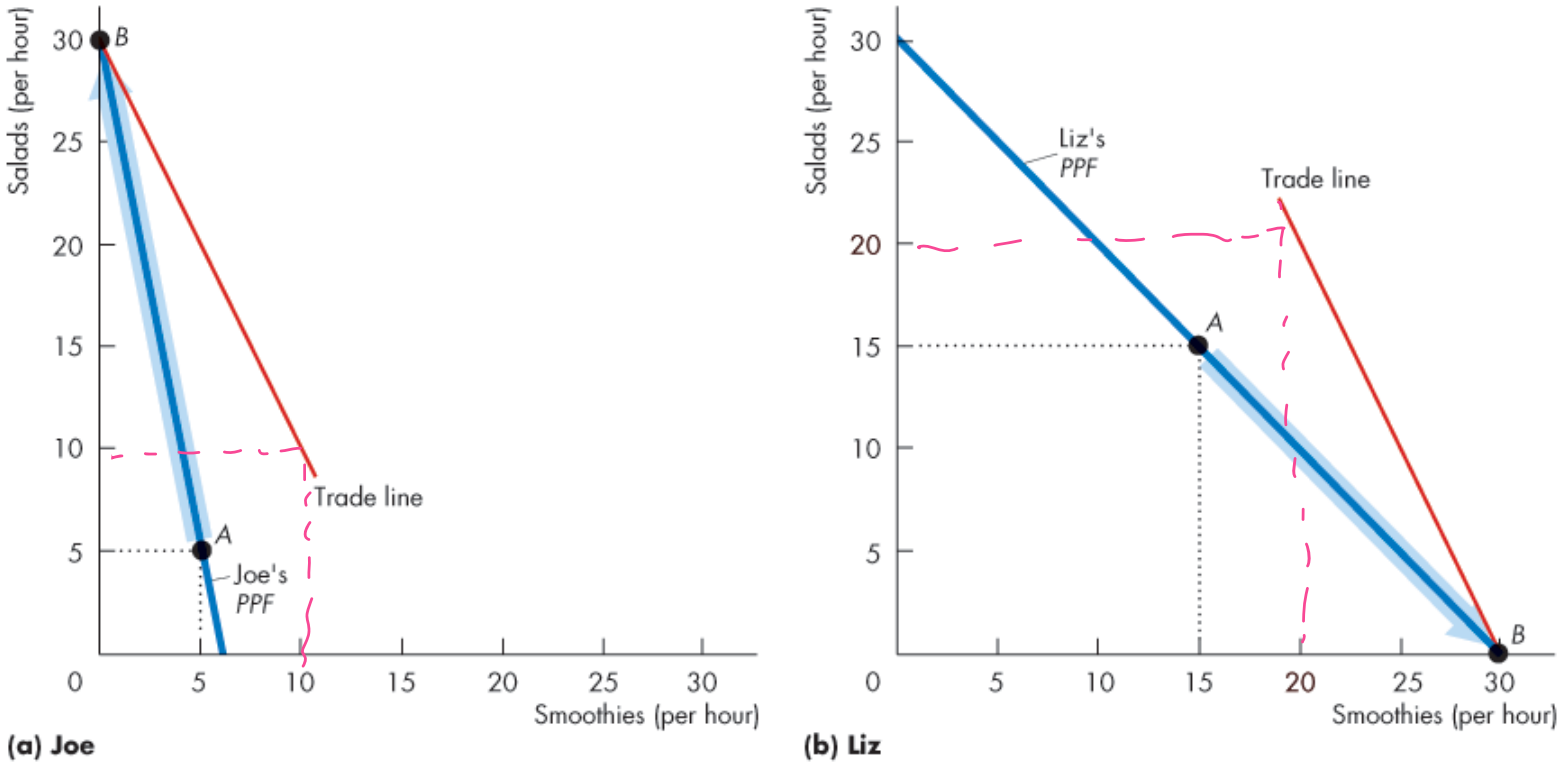


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Gains from Trade

They trade salads for smoothies along the red "Trade line."

On the trade line, the price of a salad is 2 smoothies or the price of a smoothie is ½ of a salad.

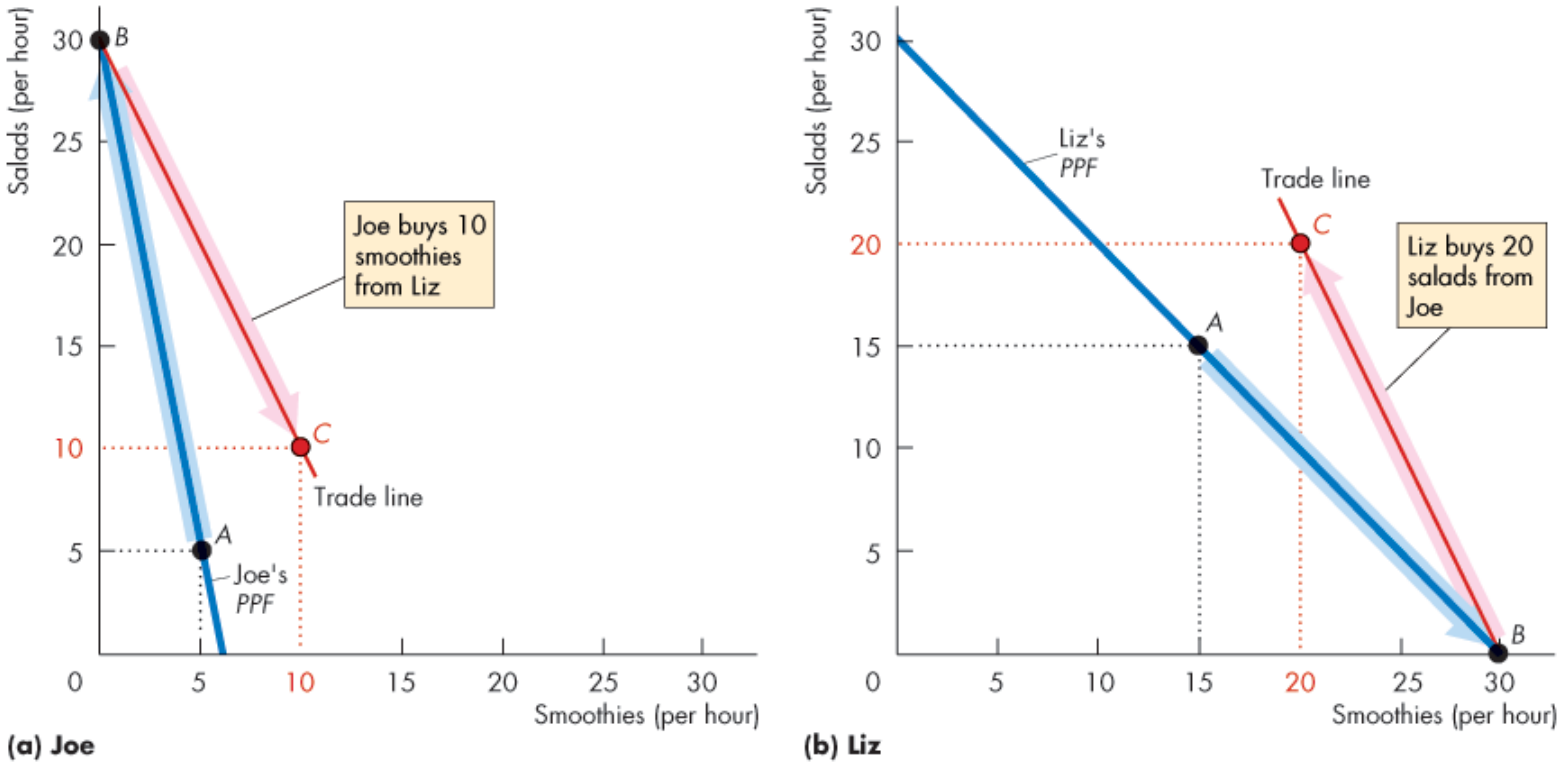


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Gains from Trade

Joe buys smoothies from Liz and moves to point *C*—a point *outside* his *PPF*.

Liz buys salads from Joe and moves to point *C*—a point *outside* her *PPF*.



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Gains from Trade

The Liz-Joe Economy and its *PPF*

With specialization and trade both Liz and Joe get outside their *PPF*s.

If Liz and Joe are the only producers in the economy, what does the economy's *PPF* look like?

Figure 2.7 on the next slide shows the construction of the economy's *PPF*.

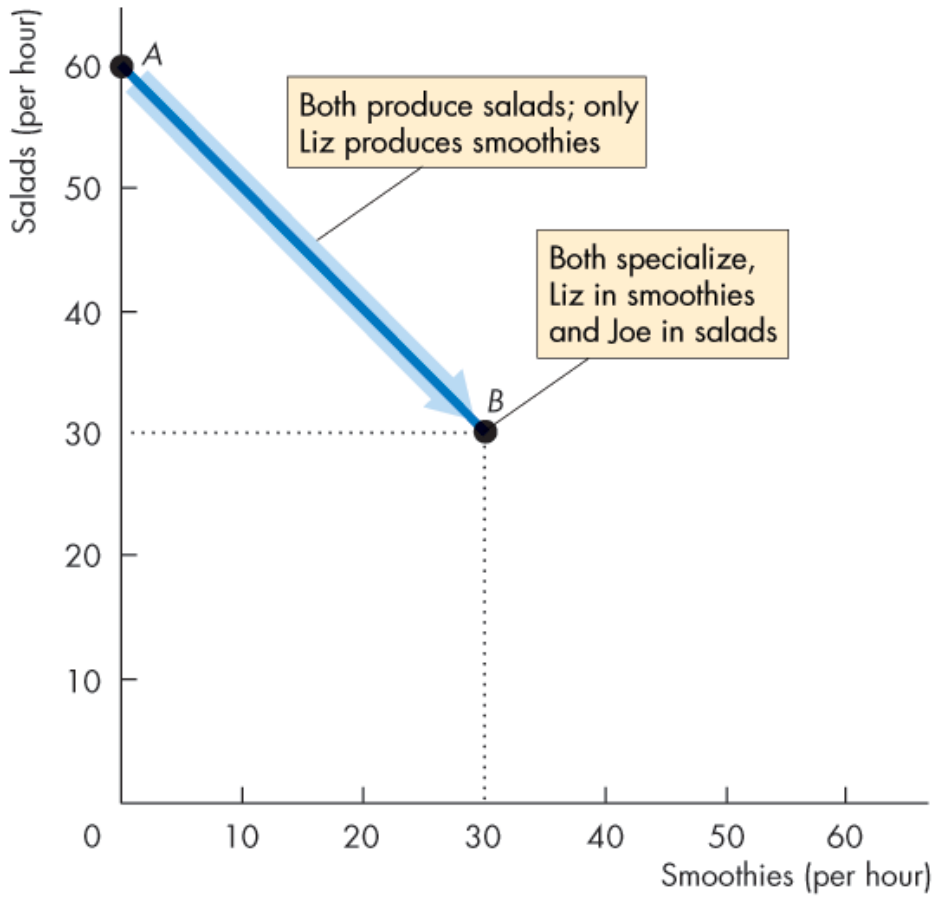
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Gains from Trade

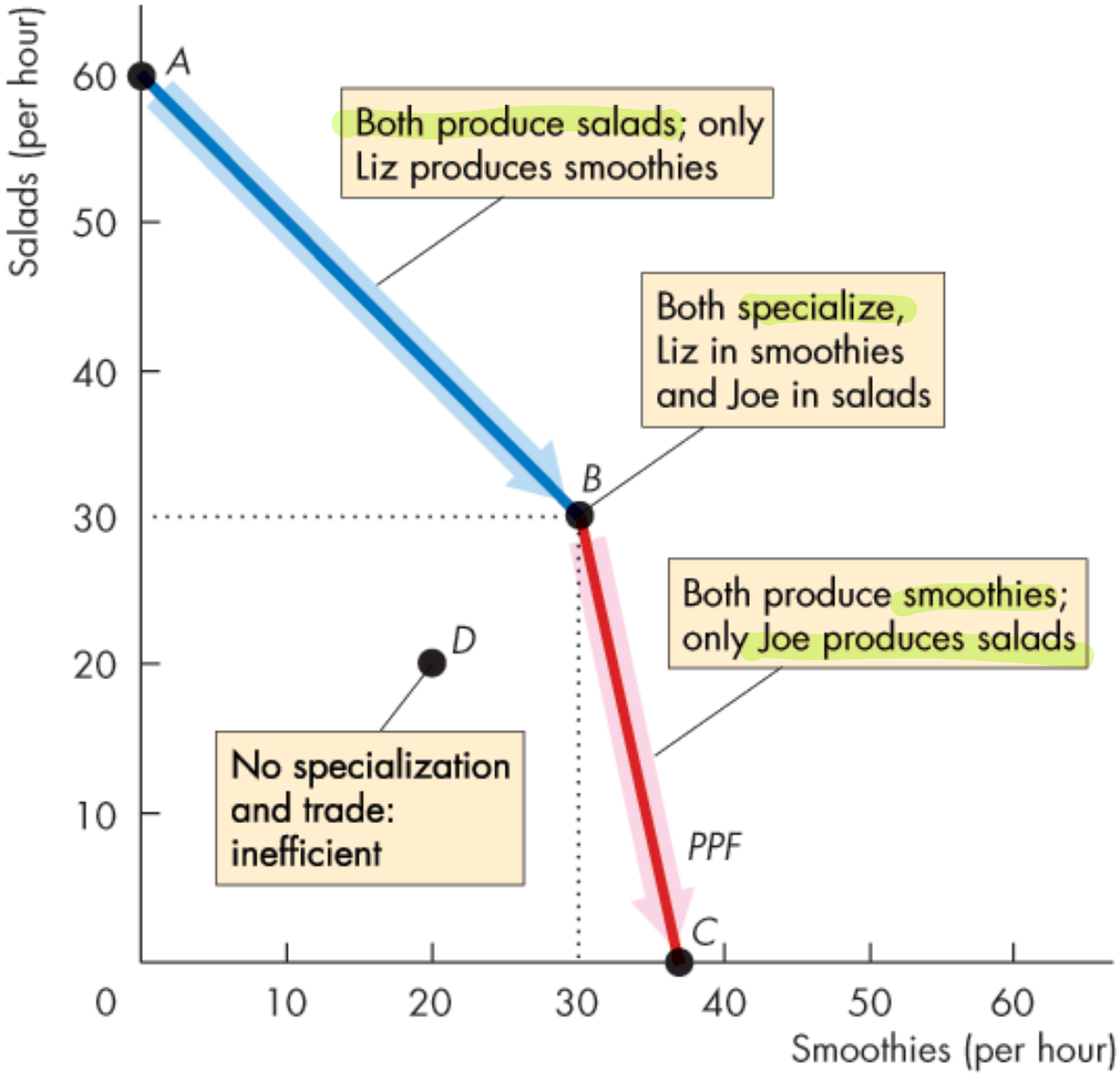
If both produce only salads, the economy produces 60 salads at point *A*.

If the economy starts to produce smoothies, Liz has the comparative advantage in smoothies and produces the first 30 smoothies at a cost of 1 salad per smoothie.

At point *B*, Liz produces 30 smoothies and Joe produces 30 salads.



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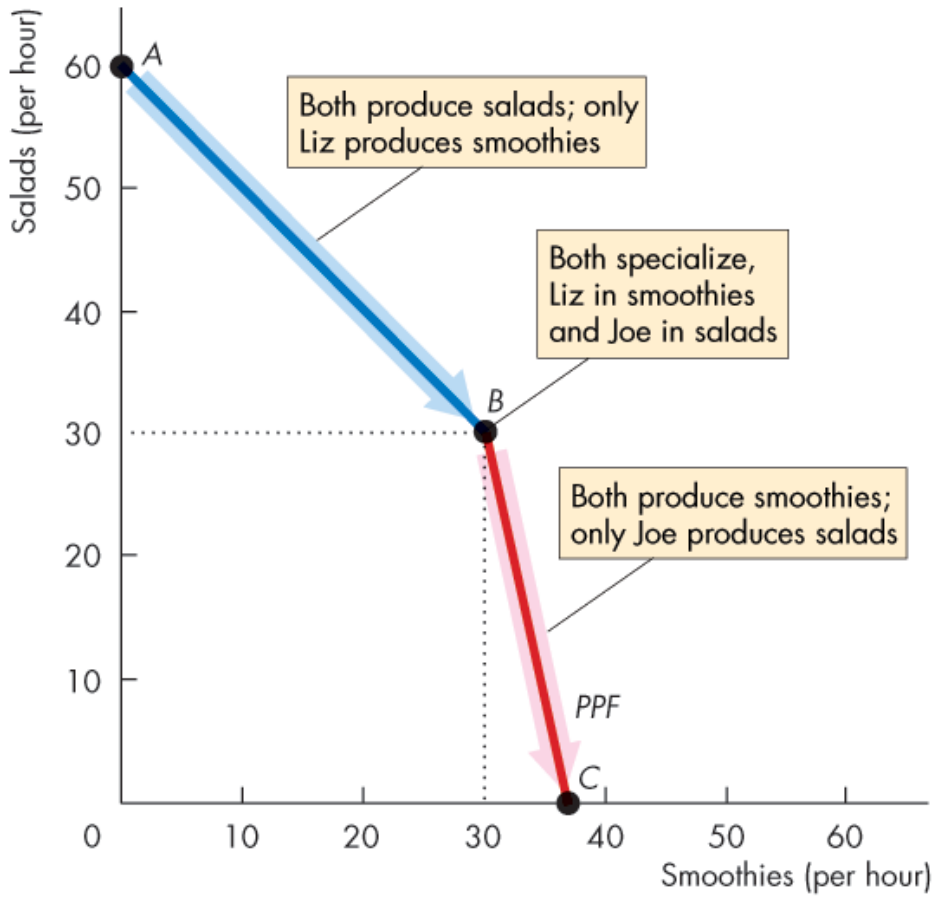
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Gains from Trade

For the economy to produce more than 30 smoothies, Joe will have to produce fewer salads and start producing smoothies.

Joe's cost of producing a smoothie is 5 salads.

If all the economy's resources are used to make smoothies, the economy produces at point *C*.



The outward-kinked curve is the Liz-Joe economy *PPF*.

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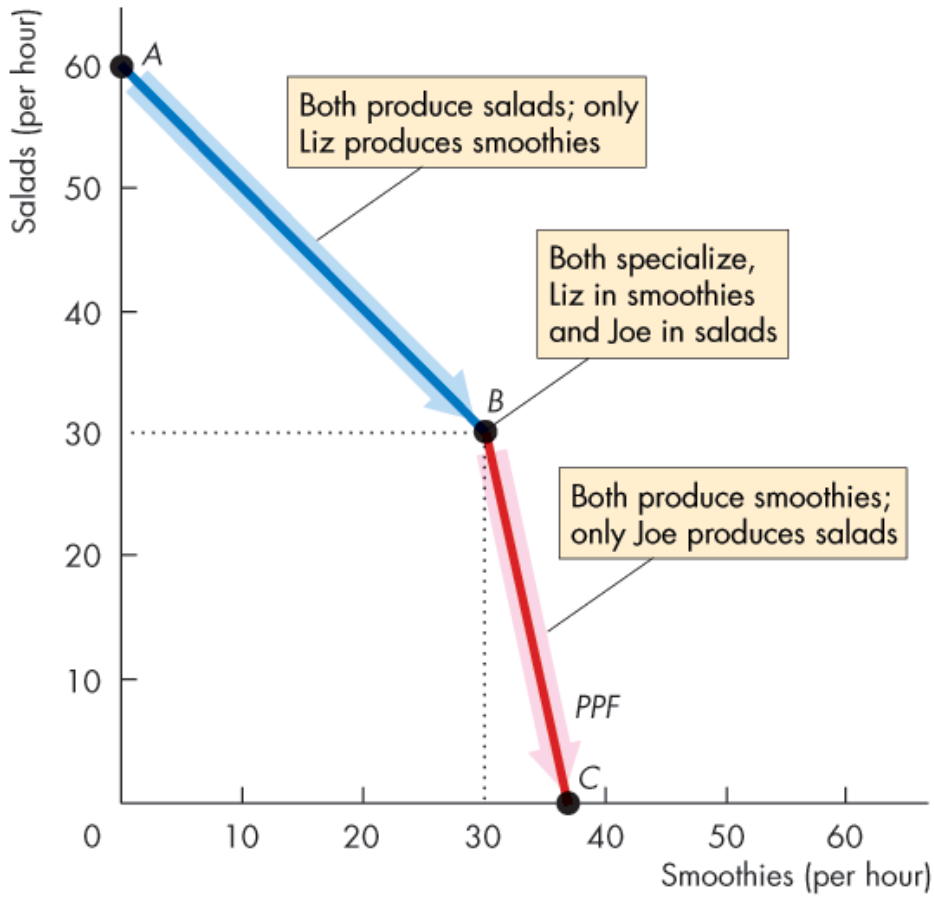
Gains from Trade

Efficiency and Inefficiency

When both Liz and Joe specialize, they produce efficiently at point *B* on the economy's *PPF*.

At all other points on the economy's *PPF*, one person specializes and production is efficient.

Production at any point on the *PPF* is efficient.

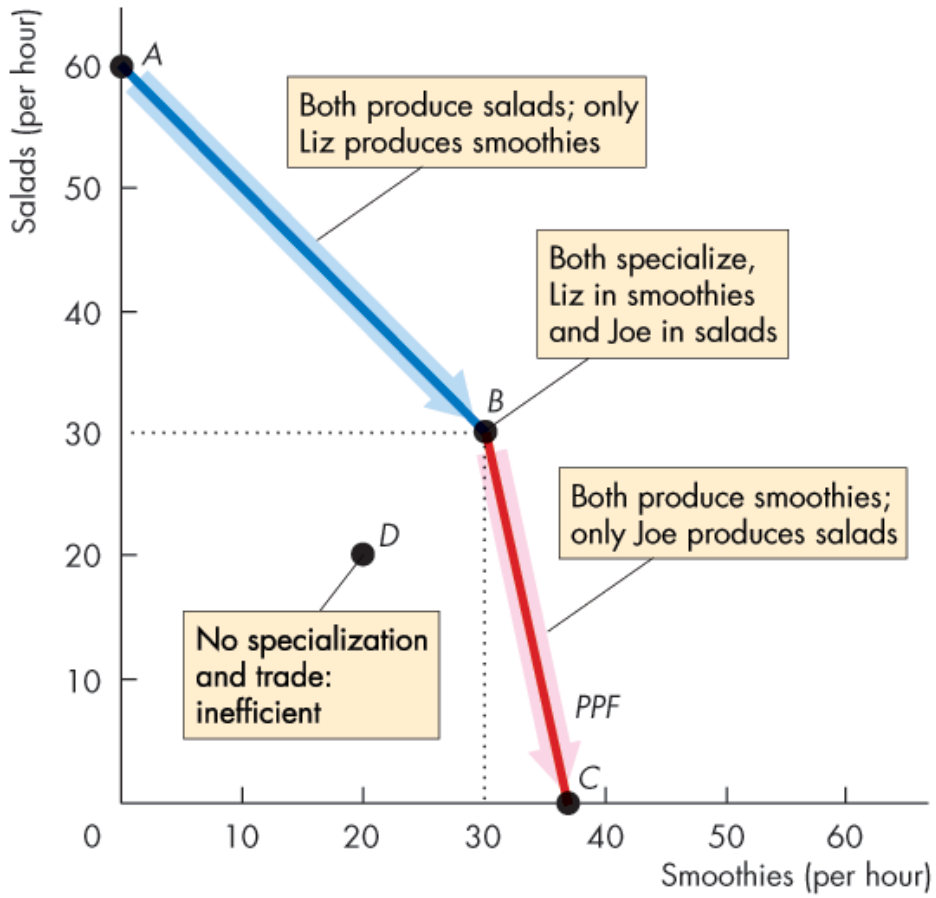


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Gains from Trade

But with no specialization, Joe and Liz produce at a point inside the economy's *PPF*.

Production at point *D* is inefficient.



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Economic Growth

more production = more growth

The expansion of production possibilities—an increase in the standard of living—is called **economic growth**.

Two key factors influence economic growth:

- Technological change
- Capital accumulation

Technological change is the development of new goods and of better ways of producing goods and services.

Capital accumulation is the growth of capital resources, which includes *human capital*.

↓ complements tech factor since tech is leveraged & utilised by humans

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Economic Growth

↑ monetary terms or exchange - for each production, comes a cost

The Cost of Economic Growth

To use resources in research and development and to produce new capital, we must decrease our production of consumption goods and services.

✓ current consumption

So economic growth is not free.

The opportunity cost of economic growth is less current consumption.

- resources such as time, electricity, money are often saved for more productive use

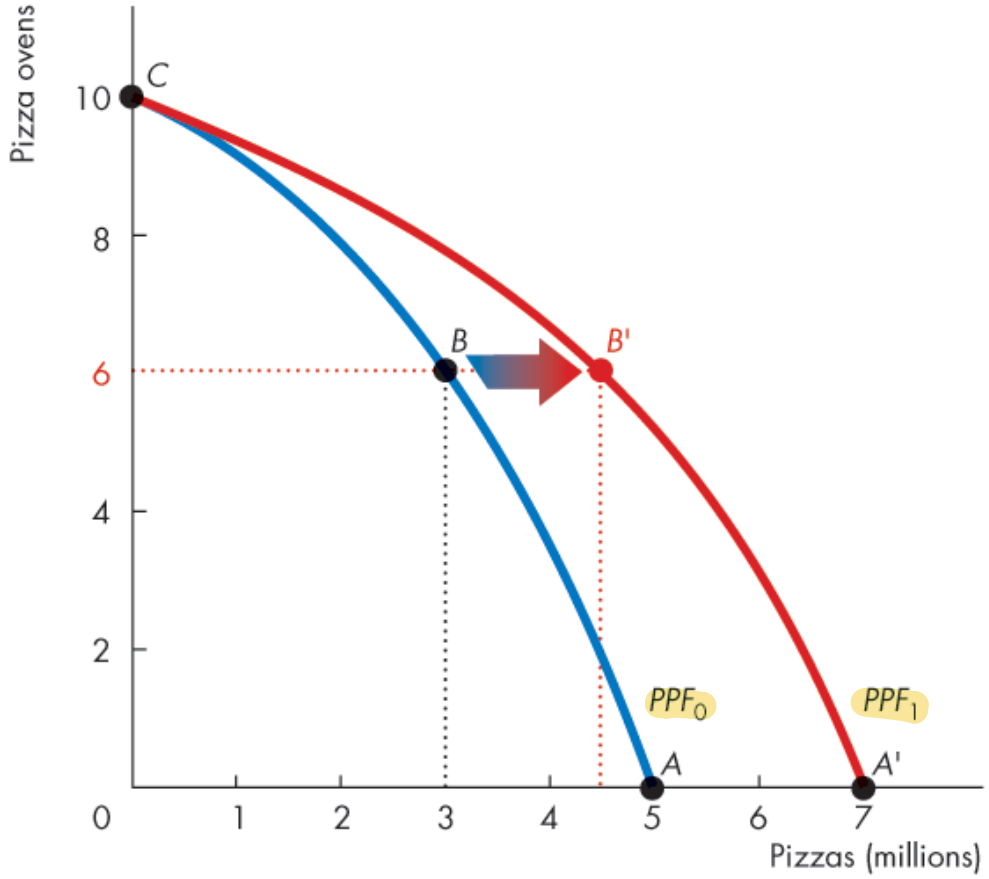
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Economic Growth

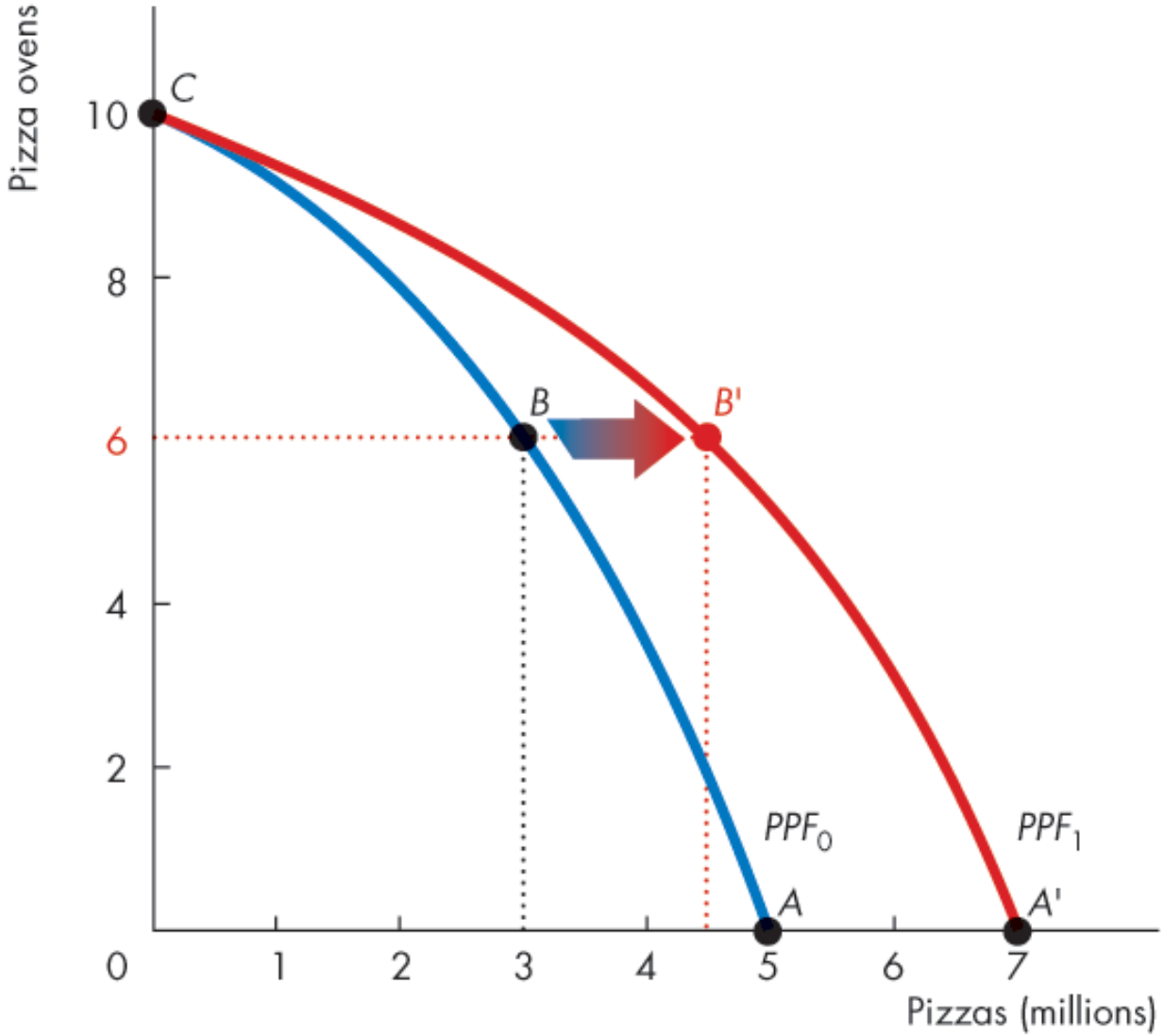
Figure 2.8 illustrates the tradeoff we face.

We can produce pizzas or pizza ovens along PPF_0.

By using some resources to produce pizza ovens today, the *PPF* shifts outward in the future.



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most important pillar

efficient stable financial systems in banking

Economic Growth

linked to:
1) high income levels
2) low unemployment rates
3) innovated tech solutions
4) high quality healthy facilities
→ high standards of living

Changes in What We Produce

always gives returns

Investment in capital and technology creates economic growth and increases income.

The model of specialization and trade explains the different patterns of production across countries.

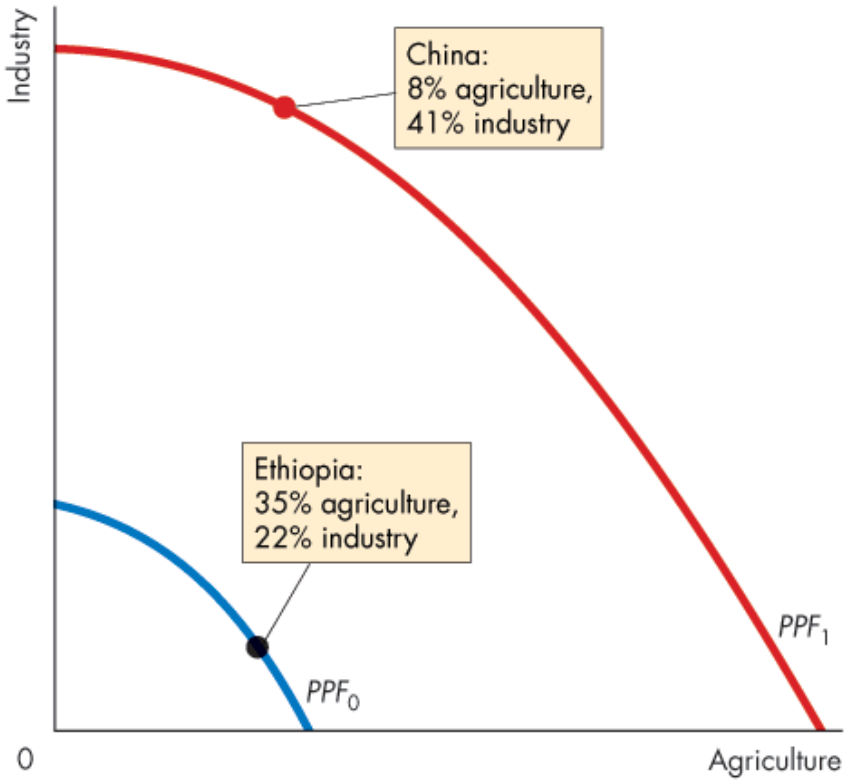
Figure 2.9 illustrates how economic growth influences the pattern of production.

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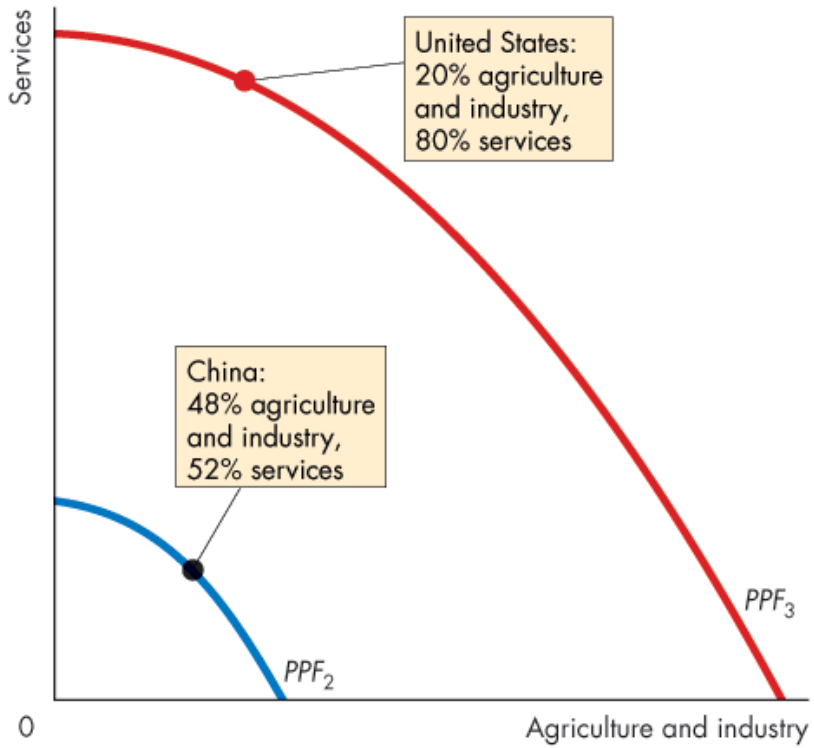
Economic Growth

Figure 2.9(a) compares low-income Ethiopia and China.

Figure 2.9(b) compares China and the rich United States.



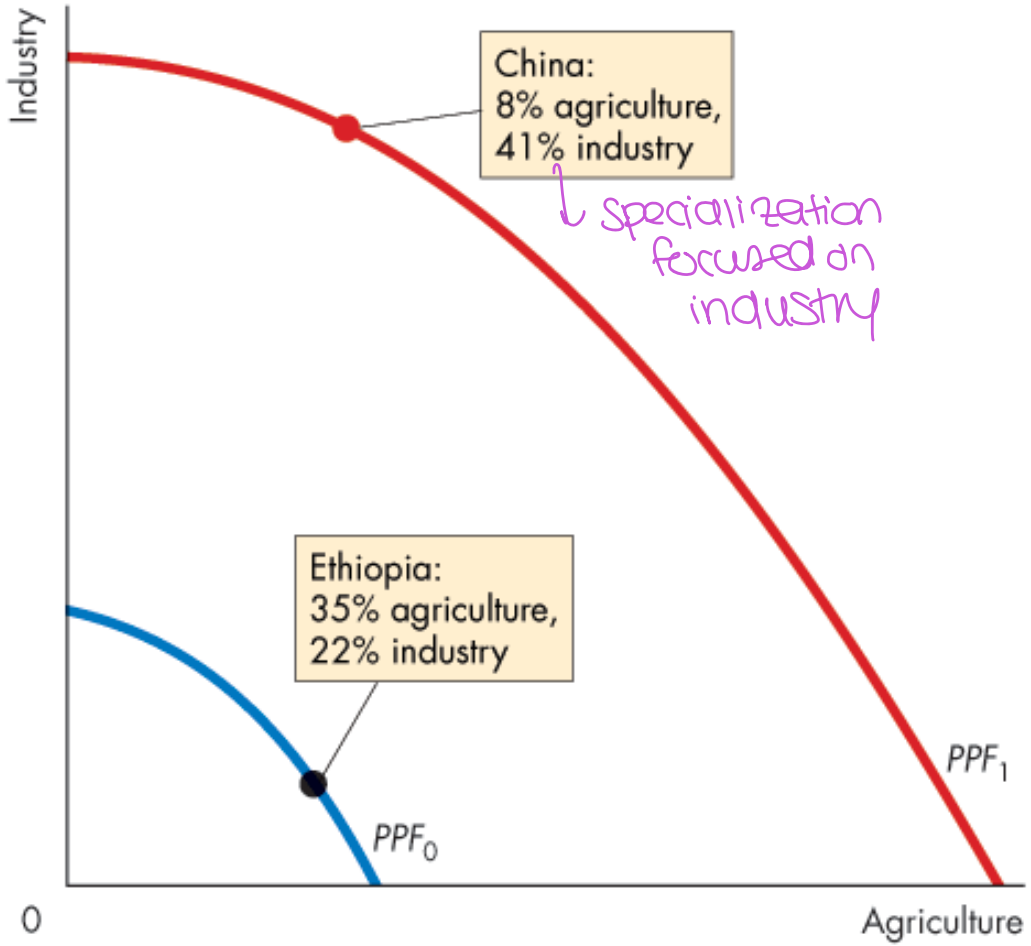
(a) From low to middle income



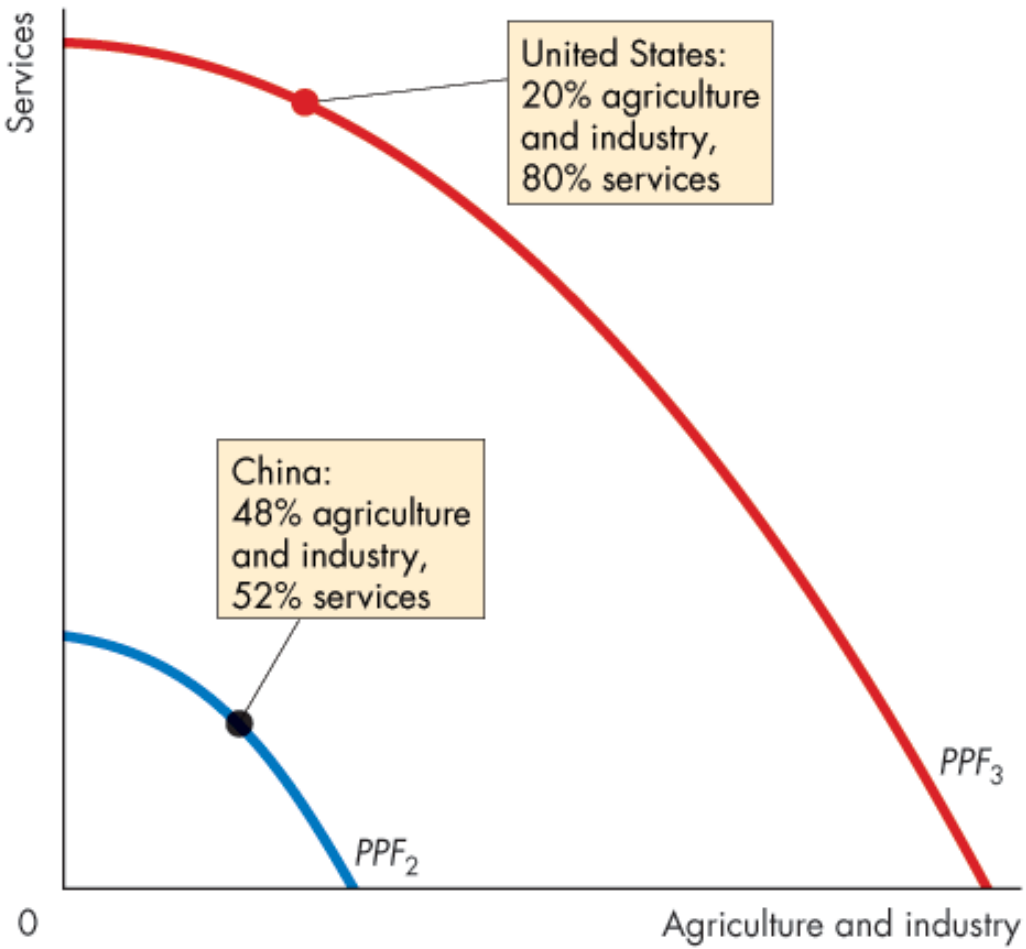
(b) From middle to high income

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→ every country has a specialization, which introduces different coordination patterns



(a) From low to middle income



(b) From middle to high income

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Economic Coordination

→ trade is not possible if decision among individuals are not well-coordinated

To reap the gains from trade, the choices of individuals must be coordinated.

To make coordination work, four complimentary social institutions have evolved over the centuries:

1. Firms
2. Markets → local / international
3. Property rights → claim of ownership
4. Money → token to exchange
   → widely accepted

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Economic Coordination

A **firm** is an economic unit that hires factors of production and organizes those factors to produce and sell goods and services.

A **market** is any arrangement that enables buyers and sellers to get information and do business with each other.

Property rights are the social arrangements that govern ownership, use, and disposal of resources, goods, or services.

Money is any commodity or token that is generally acceptable as a means of payment.

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Economic Coordination

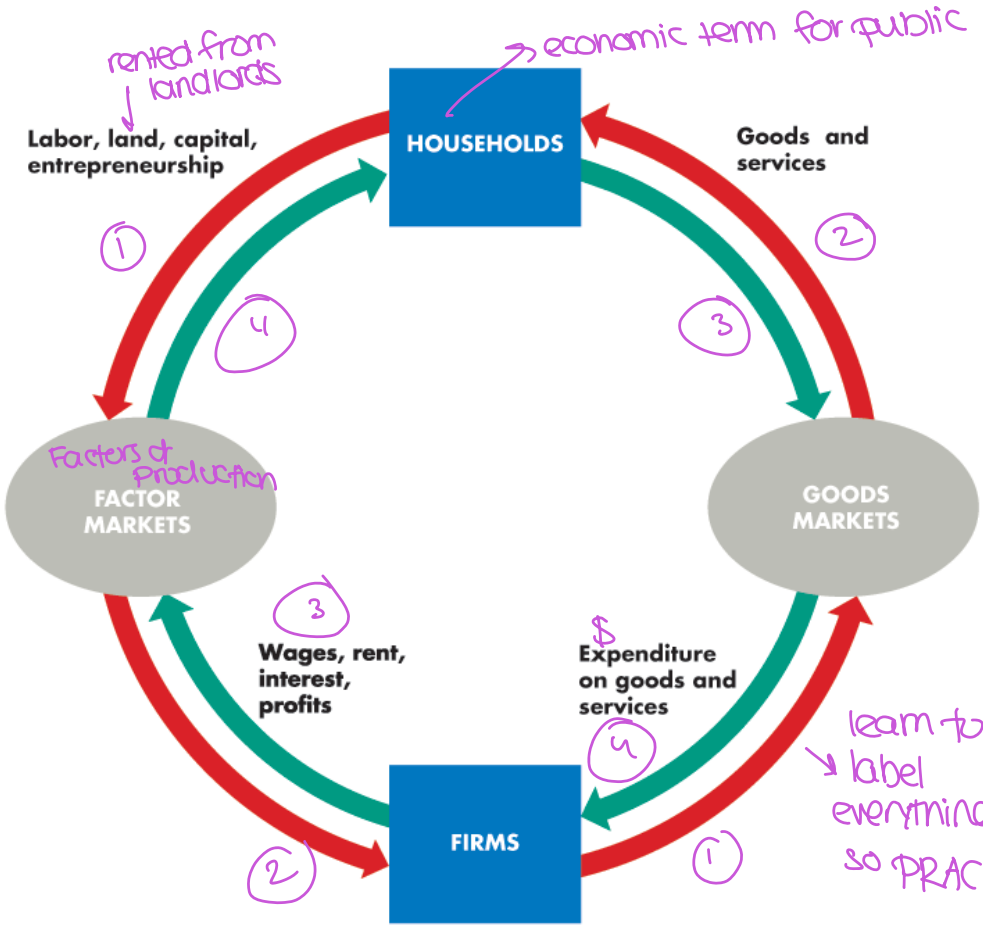
Circular Flows Through Markets

Figure 2.8 illustrates how households and firms interact in the market economy.

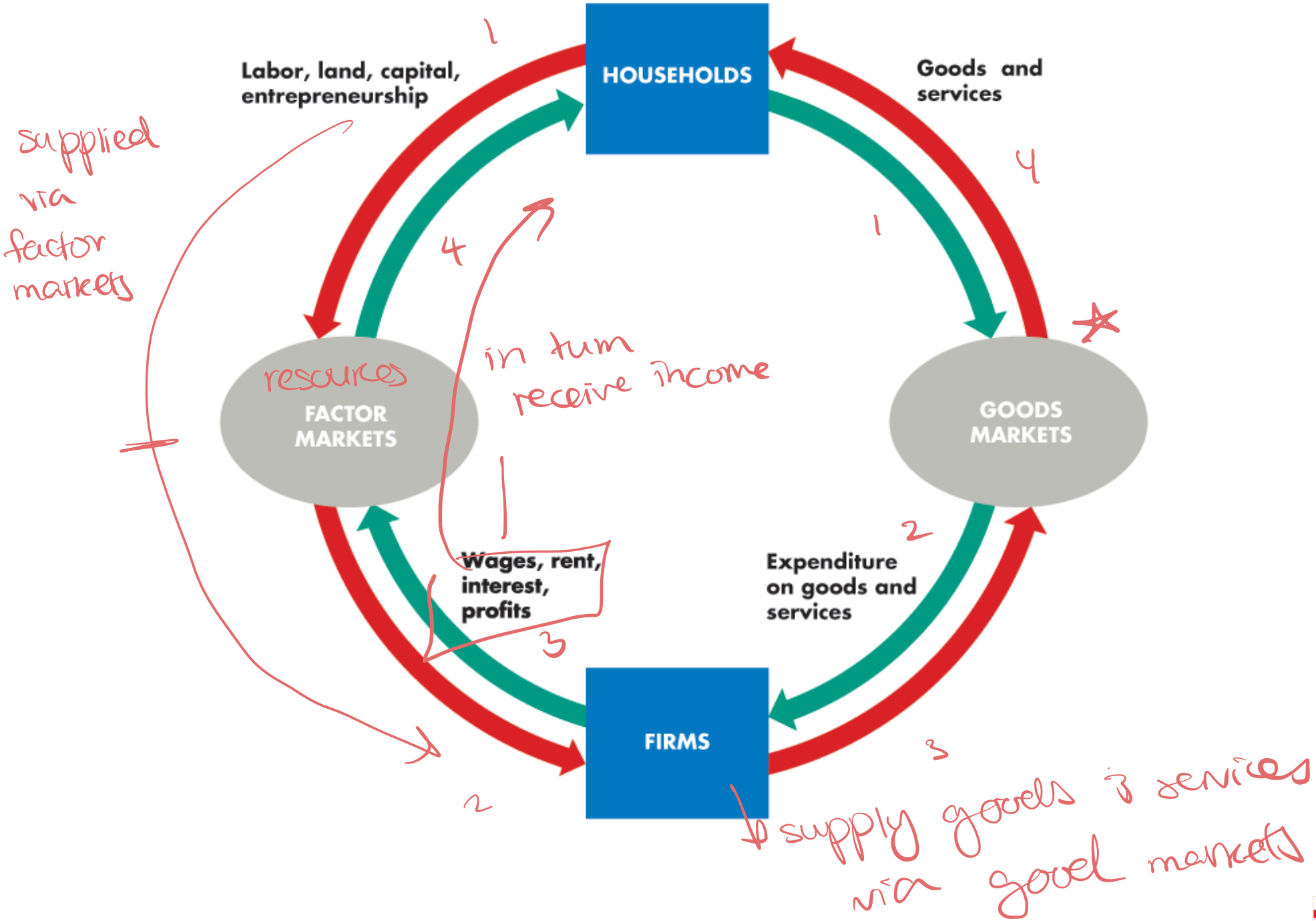
Factors of production, and …

goods and services flow in one direction.

Money flows in the opposite direction.



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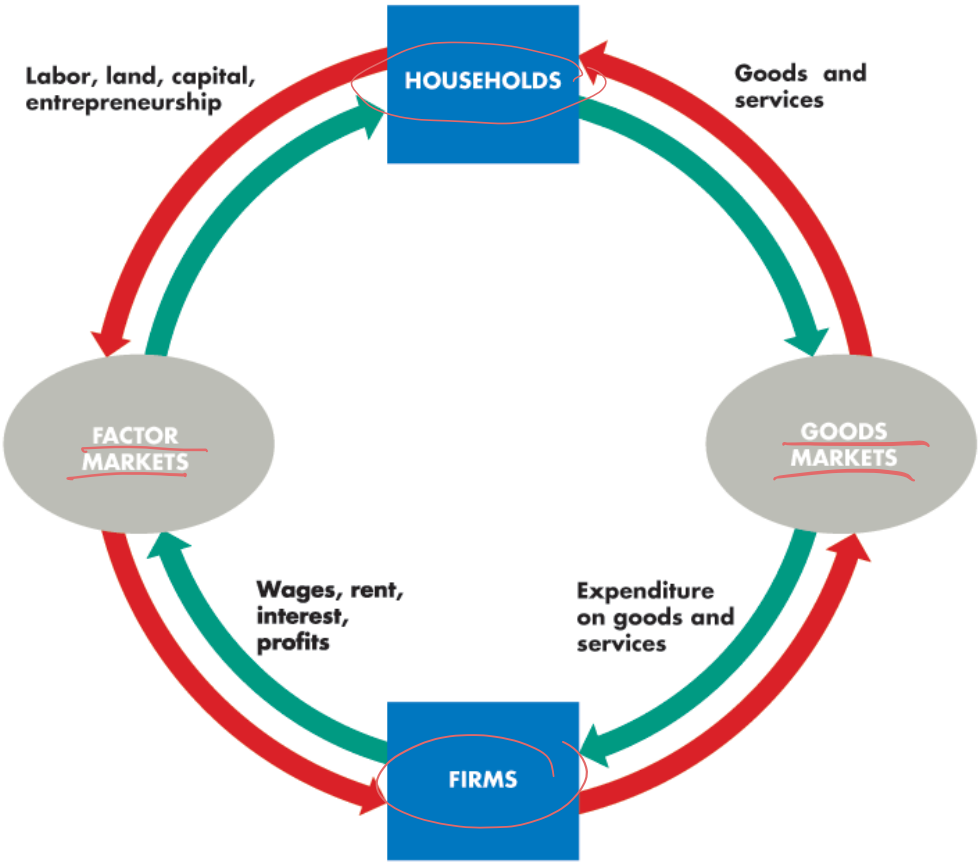


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Economic Coordination

Coordinating Decisions

Markets coordinate individual decisions through price adjustments.



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